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A SYSTEMATIC REVIEW OF HOW HEALTH LITERACY IS DEFINED AND MEASURED IN CURRENT HEALTH AND MEDICINE RESEARCH

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A SYSTEMATIC REVIEW OF HOW HEALTH LITERACY IS DEFINED AND MEASURED IN CURRENT HEALTH AND MEDICINE RESEARCH

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Abstract

Objectives: Researchers have increasingly worked to identify challenges associated with health literacy, but the way health literacy is understood (conceptualized) should be closely linked to how it is measured (operationalized) in each study context. This study aimed to gain insights into how health literacy is defined and measured in current health literacy research and examine the relationship between health literacy definitions and instruments.

Method: A systematic review was conducted to identify research studies on health literacy. The Medline, PsycINFO, ERIC, and CINAHL databases were searched for articles published in two randomly selected months in 2019. All health literacy definitions and instruments were considered in relation to two predefined categories inspired by Nutbeam's descriptions of different health literary levels.

Result: We found a high number of health literacy research articles, within both public health and the clinical field, that utilized a large variety of instruments and definitions. The majority of the studies provided a health literacy definition. However, in almost half of these studies, health literacy was defined in a broad perspective (including aspects such as social health literacy competence and the ability to process and appraise health information) but used instruments with a narrower focus (measuring basic skills and knowledge).

Discussion/conclusion: Due to the high degree of inconsistency between health literacy definitions and instruments in current health literacy research, there is a risk of missing important information about health literacy that was argued to be important in studies' initial understanding of the concept. This gap should be considered in future health literacy research.

PROSPERO registration number: CRD42020179699

Strengths and Limitations of This Study

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- To our knowledge, this is the first systematic review to investigate connections between health literacy definitions and instruments used in current health literacy research
- The health literacy definitions and instruments were categorized based on an established health literacy theory described by Nutbeam
- The initial plan of assessing health literacy definitions and instruments for three categories was changed to two due to difficulties in distinguishing between two of the categories
- Searches were limited to two randomly selected months

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Introduction

Health literacy is usually understood as cognitive and social skills which determine the motivation and ability to understand and use health information,[1] and adequate health literacy is seen as a prerequisite for healthy behaviours. Researchers have increasingly worked to identify challenges associated with health literacy and investigate the role they play in an individual's ability to comprehend self-care information and its relationship to health outcomes.[2] Empirical studies have reported that low health literacy is associated with poor health-related outcomes, such as high hospital admission rates,[3-4] low participation in preventive activities,[5-9] poor self-management of chronic conditions,[9] poor disease outcomes, and high mortality.[11-13]

The concept of health literacy emerged in the 1970s when health education was viewed as social policy.[14] A topic-specific query in the PubMed tools reveals an exponential growth of articles about health literacy, with 129 references between 1986 and 1990 compared to more than 8000 in the last five years. Today, health literacy is seen as a global goal to enhance health promotion through improved education and communication strategies to improve health outcomes.[15]

Health literacy is defined in numerous ways.[14, 16-18] In a systematic review by Sorensen et al.,[14] 17 different definitions and 12 conceptual models were reported on health literacy. Another systematic review found that 34 research articles between 2010 and 2015 had an explicit objective to define the concept of health literacy.[17] Moreover, the literature has reported the use and development of more than 150 health literacy instruments over the last decade.[19-21] Traditionally, health literacy approaches have focused on individual skill deficits and health education based on the communication of factual information on health risks as well as how to use the health system[22] while the majority of health literacy research has used

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instruments measuring reading and numeric skills.[23] However, in recent years, more multi-dimensional perspectives and instruments measuring health literacy have been introduced, such as the Health Literacy Questionnaire[24] and the Health Literacy Survey European Questionnaire 47.[25]

The current and sometimes confusing use of various and inconsistent interpretations of health literacy is a challenge for the development of valid and reliable measurements.[16] According to Nutbeam,[26] health literacy (HL) contains three levels, progressing from basic skills in reading and writing (functional HL) to the ability to derive meaning from different forms of communication and apply new information to changing situations (interactive HL) and to more advanced cognitive skills which, together with social skills, can be applied to critically analyse information the ability to achieve policy and organizational changes (critical HL).

Different understandings of health literacy and different measurement tools may be useful as they complement each other and provide different perspectives. However, the way health literacy is understood (conceptualized) should be closely linked to how it is measured (operationalized) in each study context.[23] Ngyen et al. has described this as a "conceptual stumbling block" that needs to be resolved for the field to progress.[23] A first useful step for addressing this might be to systematically explore how it appears in current health literacy research. Hence, by performing a systematic review, our aim is to gain insights into how health literacy is defined and measured in current health literacy research. In particular, we will examine the relationship between health literacy definitions and instruments. This review may increase our understanding of potential conceptual and methodological challenges or gaps that need to be addressed in future research.

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Methods

This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement[27] and registered in PROSPERO (No. CRD42020179699): https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=179699. The review was designed with a time frame limited to two randomly selected months in 2019. Due to a high number of health literacy articles published every year, random selection was chosen in order to reflect current health literacy research.

Search Strategy

Systematic literature searches were conducted in collaboration with a trained librarian (3.4.2020). The Medline, PsycINFO, ERIC, and CINAHL databases were searched for the term "health literacy". For all databases except for ERIC (where this was not possible), the search was automatically restricted to two randomly selected months in 2019: March and October. Citations in ERIC were manually assessed for articles published in March 2019 and October 2019

The search yielded a total number of 1031 citations. Endnote™ Version X9 was used to manage the generated research articles. After removing duplicates, 951 citations remained. All records not published for the first time in 2019 were removed, leaving 414 articles (see Appendix 1 for the search history).

Selection Criteria

The inclusion and exclusion criteria were developed a priori. The 414 published articles were distributed among six researchers (KHU, AKW, MHA, CB, SH, MHL), who worked in pairs.

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The articles were included if it they fulfilled the following inclusion criteria: 1) "health literacy" was mentioned in the title or abstract, 2) the article was peer reviewed, 3) the research was original, 4) it used a quantitative design, 5) it was published in English language, and 6) the study population included individuals over 16 years of age. The researchers independently screened the articles for eligibility according to the criteria and then met in pairs to compare their results, resolve any conflicting opinions, and decide whether to include each article. Conflicting opinions in pairs were presented and solved by the whole team.

Data Extraction (Selection and Coding)

To achieve consistency in the data extracted from the included articles, an Excel spreadsheet was created. The initial question in this spreadsheet was whether health literacy was measured. If the answer was no, the article was excluded. The data extracted from the articles included information about the study design and context, such as country of origin and whether the study was conducted in a public health or clinical health setting. For clinical studies that included participants with health problems, the different types of diseases were categorized according to the International Statistical Classification of Diseases and Related Health Problems (ICD, WHO, 2010).[28] Furthermore, data included information about the health literacy instrument utilized—namely, whether it was generic or disease specific, if it was used in combination with other health literacy instruments, and whether a reference was provided for the instrument.

The published articles were analysed regarding which health literacy definitions and instruments were used. Before the data extraction, a pilot assessment of three studies was performed to determine the feasibility of the data extraction tool. The initial plan was to assess health literacy definitions and instruments for three categories 1) functional, 2) interactive, and 3) critical health literacy which were inspired by Nutbeam's description of

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different health literacy levels 26. As a result of the pilot testing, the number of categories was changed due to difficulties in distinguishing between the two latter levels. All health literacy definitions and health literacy instruments were therefore considered in relation to two categories; category 1 included definitions and instruments describing basic reading and writing skills, disease-specific knowledge, and practical skill competencies needed to function in everyday situations and category 2 included definitions and instruments that also described health literacy as skills to communicate and interact with healthcare providers as well as the ability to interpret and critically analyse health information. (Appendix 2 illustrates the coding for the data extractions).

The researchers first extracted data and considered definitions and instruments related to the two categories independently. They then met in pairs to compare their results and resolve any conflicting opinions. The remaining conflicting opinions within pairs were presented and solved by the whole team in a group meeting.

Patient and public involvement

No patient involved

Results

The search yielded 1031 articles in total (see the Appendix for the search history). After removing duplicates, 951 articles remained. Of the 163 articles that were read in full texts, one was a duplicate and 39 were excluded because they did not measure health literacy. Ultimately, 120 articles were included (see Figure 1).

Figure 1 about here

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Characteristics of Included Studies

Sixty articles described studies with a clinical health focus, and 60 were conducted in public health settings. The clinical studies included a great variety of diseases, but the most frequent diseases were those in the circulatory system (n=11) and endocrine diseases (n=8).

The majority of the studies utilized a cross-sectional design (n=90). Nine studies utilized a randomized controlled design, and 10 studies were quasi-experimental. Four of the studies used a longitudinal design while one was a case-control study. Figure 2 presents an overview of all study designs.

Figure 2 about here

As Figure 3 indicates, the majority of studies were conducted in North America (n=56) and Asia (n=31). The fewest studies were performed in Africa (n=4) and South America (n=3).

Figure 3 about here

Health Literacy Definitions and Instruments

Out of the 120 included articles, 77 utilized instruments from Category 1 (describing basic reading and writing skills, disease-specific knowledge, and practical skills), whereas 43 instruments were from Category 2 (described communication and interaction skills and/or the ability to interpret and critically analyse health information). The most frequently used HL

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instruments in Category 1 were Newest Vital Sign[29] (n=19) and Test of Functional Health Literacy [30] (n=13). The most frequently used HL instruments in Category 2 were The European Health Literacy Survey Questionnaire[31] and The eHealth Literacy Scale[32] (n=10).

Of the 120 articles included, 80 provided a health literacy definition; eight of these definitions were from Category 1 and 72 were from Category 2. The most frequently used reference for Category 2 health literacy definitions was Nutbeam.[33-35] The most frequently used reference for Category 1 definitions was the definition provided by Jorm and colleagues.[36-37] Table 1 gives an overview of the text in the most cited health literacy definitions. Words in italics are examples of key words considered important in the decision to adding the definition to Category 1 or Category 2.

Table 1. The most frequently used references for health literacy definitions in the included studies

Reference	**Definition**	**Number of studies**
Category 2		
Nutbeam (1998, 2000, 2015)	"The *cognitive and social skills* which determine the motivation and ability of individuals to gain access to understand and use information in ways which promote and maintain good health."	19
Sørensen (2012)	"Health literacy entails people's knowledge, motivation and competences to access, *understand*, *appraise*, and *apply* health information in order to make judgments and take decisions in everyday life concerning healthcare, disease prevention and health promotion to maintain or improve quality of life during the life course."	14
The Institute of Medicine (2004)	"The individuals' capacity to *obtain, process* and understand basic health information and services needed to make appropriate health decisions."	11
Category 1		
Jorm (1997) Jorm (2012)	"knowledge about appropriate treatment options; and *attitudes* that facilitate recognition and treatment-seeking" "*knowledge and beliefs* about mental disorders which aid their recognition, management or prevention"	5
American Medical	"The *constellation of skills, including the ability* to perform basic reading and numerical tasks required to function in the healthcare environment"	1

For peer review only

Association (AMA) (1999)		

Regarding the connection between health literacy definitions and instruments in the 80 relevant studies, 41 articles used health literacy definitions and instruments characterized at the same level. Thirty-three of these were in Category 2 and 8 were in Category 1 (see Figure 4 and Table 2)

Figure 4 about here

Table 2. Studies categorized with mutual levels of HL definitions and HL instruments

Study reference	**Health literacy definition reference**	**Cat.**	**Health literacy instrument**	**Cat.**
Degan et al., 2019	World Health Organization, 2009	2	The Health Literacy Questionnaire (Osborn et al., 2013)	2
Ernsting et al., 2019	Sørenesen, 2012	2	European Health Literacy Survey Questionnaire - short form (Sørensen et al., 2013) and The eHealth Literacy Scale (Norman, 2006)	2
Evans et al., 2019	Sørenesen, 2012	2	European Health Literacy Survey Questionnaire (Sørensen et al., 2013)	2
Goto et al., 2019	Nutbeam, 1998	2	The Functional Communicative and Critical HL Tool (Ishikawa et al., 2008)	2
Guclu et al., 2019	World Health Organization, 2009	2	European Health Literacy Survey Questionnaire (Sørensen et al., 2013)	2
Kobayashi et al., 2019	Sørensen, 2012	2	European Health Literacy Survey Questionnaire (Sørensen et al., 2013)	2

For peer

Lim et al., 2019	Osborne, 2013	2	Health Literacy Questionnaire (Osborne et al., 20139	2
Logullo et al., 2019	Sørensen, 2012	2	Short Assessment of Health Literacy for Portuguese Speaking Adults (SAHL-PA) (Apolonario et al., 2012)	2
Lu et al., 2019	Sørensen, 2012	2	European Health Literacy Survey Questionnaire -HLS EU-16 (Sørensen et al., 2013)	2
Gûner et al., 2019	Nutbeam, 2000	2	Questionnaire developed within the study context (consisting of 21 questions about HL knowledge, practice, and attitudes)	2
Indino et al 2019	Nutbeam, 2015	2	The Functional Communicative and Critical Health Literacy Tool (Ishikawa et al., 2008)	2
Oh et al., 2019	World Health Organization, 2009	2	The eHealth Literacy Scale (Norman, 2006)	2
Oliffe et al, 2019	Peerson and Saunders, 2009, O'Brian et al., 2005	2	The Health Literacy Questionnaire (Osborn et al., 2013)	2
Rababah et al, 2019	World Health Organization, 2013, Sørensen, 2012	2	The Health Literacy Questionnaire (Osborn et al., 2013)	2
Zhang et al., 2019	Sørensen, 2012	2	Heart Failure-specific Health Literacy scale (HFHL) (Matsuka et al., 2018)	2
Uysal et al., 2019	Sørensen, 2012	2	European Health Literacy Survey Questionnaire (Sørensen et al., 2013)	2
Lu et al., 2019	Sørensen et al., 2012	2	European Health Literacy Survey Questionnaire -HLS EU-16 (Sørensen et al., 2013)	2
Brandsetter et al., 2019	Sørensen, 2012	2	The European Health Literacy Survey (HLS-EU) (Sørensen et al., 2013)	2
Wang et al., 2019	Kutner et al., 2003	2	Chinese Health Literacy Scale for Diabetes (Leung, 2013)	2
Ruëgg and Abel, 2019	Selden et al., 2000	2	Short survey tool for public health and health promotion research (Abel, 2014)	2
Strømer et al., 2019	World Health Organization, 1998	2	The Health Literacy Questionnaire (Osborn et al., 2013)	2
An et al., 2019	Nutbeam, 2000	2	The Health Literacy Scale (Chew, 2004)	2
Erunal et al., 2019	Nutbeam, 2000	2	Turkish Health Literacy Scale -32 (Okyay et al., 2016)	2

Eo et al., 2019	Nutbeam, 2000	2	The Health Literacy Assessment Scale for Asian Immigrant Women (And and Yang, 2015)	2
Fernandez Gutierrez et al., 2019	Nutbeam, 2000	2	European Health Literacy Survey Questionnaire -HLS EU-16 (Sørensen et al., 2013) Health Literacy App Questionnaire + five practical tests	2
Hu et al., 2019	Nutbeam, 2000	2	Health Literacy Management Scale (Jordan, 2013)	2
Solhjoo Niloofar et al., 2019	Sørensen, 2012	2	The eHealth Literacy Scale (Norman, 2006)	2
Indino et al., 2019	Nutbeam, 2015	2	The Functional Communicative and Critical HL Tool (Ishikawa et al., 2008)	2
Wang et al., 2019	Nutbeam, 2009	2	Diabetes Health Literacy Scale (Lee, 2018)	2
Pobhirun and Pinitsoontorn, 2019	Nutbeam, 1998	2	Questionnaire developed within the study context including 6 dimensions: cognitive skills, access, communication skills, self-management, media literacy and decision-making skills	2
Nokes et al, 2019	Norman et al., 2006 and 2011	2	The eHealth Literacy Scale (Norman, 2006)	2
Kaper et. al, 2019	Kwan, 2006	2	Questionnaire developed within the study context by combining subscales from other questionnaires	2
Ma et al., 2019	Norman et al., 2006	2	The eHealth Literacy Scale (Norman, 2006)	2
Joncho et al., 2019	Jorm et al., 1997	1	Developed within the study context: Mental Health Literacy Questionnaire	1
Nye et al., 2019	U.S. Department of Health and Human Services, 2010	1	The Upper Peninsula Oral Health Assessment Survey	1
Clough et al., 2019	Jorm et al., 1997	1	Mental Health Literacy Scale (O'Connor and Casey, 2015)	1
Williston et al., 2020	Jorm et al., 1997	1	Mental Health Literacy Scale (O'Connor and Casey, 2015)	1
Waldmann et al.,2020	Jorm et al., 2012	1	Mental Health Knowledge Schedule. Depression Literacy Scale (Wilson, 2015)	1
Cehuen Neto et al., 2019	American Medical Association (AMA) ad hoc committee on health, 1999	1	Short Test of Functional Health Literacy in Adults (Parker, 1995)	1
Clough et al., 2019	Jorm et al. 1997	1	Mental Health Literacy Scale (O'Connor & Casey, 2015)	1

Lichtveld et al, 2019	Finn and O'Fallon, 2019 Gray, 2018	1	Developing and validating a questionnaire measuring environmental health literacy: Measuring media-specific knowledge, attitudes, and behaviors	1

In the remaining 39 studies, there was a disconnect between levels of health literacy definitions and instruments. In all of these, health literacy definitions were from Category 2 and all instruments were in Category 1 (see Table 3).

Table. 3 Studies categorized with conflicting levels of HL definitions and HL instruments

Study reference	**Health literacy definition reference**	**Cat.**	**Health literacy instrument**	**Cat.**
Avci et al.,2019	US Department of Health and Human Services, 2000	2	Newest Vital Sign (NVS) (Weiss et al., 2005) and Rapid Estimate of Adult Literacy in Medicine (Davis, 1993)	1
Clough et al., 2019	Kutcher et al., 2016	2	Mental Health Literacy Scale (O'Connor and Casey, 2015)	1
Flynn et al., 2019	American Dental Association Council, 2015	2	Oral Health Literacy Adults Questionnaire (Sistani et al., 2014)	1
Gaikwad, 2019	Ratzan and Parker, 2000	2	Rapid Estimation of Adult Literacy in Dentistry 30-word version (Lee et al., 2007)	1
Han et al.,, 2019	Sørensen, 2012, Nutbeam, 2000	2	Assessment of Health Literacy in Cancer Screening (AHL-C) (Han et al., 2014)	1
Himes et al., 2019	Institute of Medicine, US, Committee HL, 2004	2	Chew et al.'s (2004) set of brief questions	1
Kim et al., 2019	Ratzan and Parker, 2000	2	Three questions, each addressing oral, listening, and written literacies (no reference)	1
Kim et al, 2019	Institute of Medicine, US, Committee HL, 2004	2	Short Test of Functional Health Literacy in Adults (Parker, 1995)	1
Miranda, 2019	Sørensen et al., 2012	2	Rapid Estimate of Adult Literacy Dutch version (REALM-D) (Davis, 1993)	1
Mock et al., 2019	Institution of Medicine, 2004	2	Single Item Literacy Screening (SILS) (Morris, 2006), Short Test of	1

			Functional Health Literacy in Adults (S-TOFHLA) (Parker, 1995)	
Noback et al., 2019	Paasche-Orlow et al., 2007, Berkman et al., 2011, Pendlimari et al., 2012	2	Newest Vital Sign (NVS) (Weiss et al., 2005) Musculoskeletal specific literacy survey (LiMP questionnaire) developed within the study context	1
O'Conor et al., 2019	Institute of Medicine Health Literacy, 2004	2	Short Test of Functional Health Literacy in Adults (S-TOFHLA) (Parker, 1995)	1
Penaloza et al, 2019	Nutbeam, 1998 and Institute of Medicine, 2004	2	Short Assessment of Health Literacy Spanish (SAHL-S) (Lee et al., 2006)	1
Rafferty et al., 2019	Institute of Medicine, 2004 and Sørensen, 2012	2	Questions focusing on health advice and information-seeking skills, oral literacy, and print literacy	1
Mora-Pinzon et al., 2019	Institute of Medicine, 2004	2	Short Test of Functional Health Literacy in Adults (S-TOFHLA) (Parker, 1995) Rapid Estimate of Adult Literacy (REALM-D) (Davis, 1993	1
Sany et al., 2019	United States Health Resources and Services Administration	2	Test of Functional Health Literacy in Adults (TOFHLA) (Parker, 1995)	1
Meyers et al., 2019	Institute of Medicine, 2004	2	Newest Vital Sign (Weiss et al., 2005)	1
Lindahl et al., 2020	Nutbeam and Kickbusch, 2000	2	The three-item Brief Health Literacy Screen (BHLS) (Cavanaugh et al., 2015)	1
Kino et al., 2020	Inst of Medicine, 2004	2	Threee indicators of HL (Haun et al., 2014)	1
Zhong et al., 2020	Kindig, 2004	2	Newest Vital Sign (NVS) (Weiss et al., 2005)	1
Winokur et al., 2019	Morrison et al., 2013	2	Newest Vital Sign (NVS) (Weiss et al., 2005)	1
Williams et al, 2019	Institute of Medicine, 2004	2	Newest Vital Sign (NVS) (Weiss et al., 2005)	1
Weaver et al., 2019	Ratzan and Parker, 2000 Nutbeam, 2000	2	Rapid Estimate of Adult Literacy (REALM) (Davis, 1993)	1
Bonaccorsi et al., 2019	Sørensen, 2012	2	Newest Vital Sign (NVS) (Weiss et al., 2005)	1
Carducci et al., 2019	Ratzan and Parker, 2000 Nutbeam, 2000	2	Test of Functional Health Literacy in Adults (TOFHLA) (Parker, 1995)	1
Schrivner et al., 2019	Nutbeam, 2008	2	Three questions assessing health literacy	1

Tucker et al., 2019	Ratzan and Parker, 2000	2	Newest Vital Sign (NVS) (Weiss et al., 2005)	1
Van Wormer et al., 2019	Institute of Medicine, 2004	2	Oral Health Literacy in Adults Questionnaire (Sistani et al., 2013)	1
Chen et al., 2019	Nutbeam, 2000	2	Health Literacy Scale for Diabetes (Lee et al., 2016)	1
Gûner et al., 2019	Nutbeam, 2000	2	Developed within the context of the study	1
Lin et al., 2019	Nutbeam, 2000	2	Mandarin Health Literacy Scale (Lee et al., 2012)	1
Bonaccorsi et al., 2019	Sørensen, 2012	2	Newest Vital Sign (Weiss et al., 2005)	1
Mayer et al., 2019	Nutbeam, 1998	2	Developed within the context of the study	1
Anderson et al., 2019	Defined within the study context (no reference provided)	2	Short Test of Functional Health Literacy in Adults (S-TOFHLA) (Parker, 1995)	1
Irvin et al., 2019	Peters et al., 2012	2	The Water Environmental Literacy Level Scale (WELLS) (Irvin et al., 2019)	1
Mackert et al., 2019	Berkman et al., 2010	2	Newest Vital Sign (NVS) (Weiss et al., 2005)	1
Kaur et al., 2019	Healthy People, 2010 (Oral health)	2	TS-REALD Two-Stage Rapid Estimate of Adult Literacy (Stucky et al., 2011)	1
De Melo et al., 2019	Cavanaugh, 2011	2	Short Test of Functional Health Literacy in Adults (Parker, 1995)	1
Kim et al., 2019	Joint Committee on National Education Standards, 1995	2	Knowledge questions developed in the study context	1

Discussion

This systematic review aimed to gain insights into how current research defines and measures health literacy and, in particular, whether studies consistently used definitions and instruments. The high number of articles published in the defined timeframe shows that health literacy is of high research interest in both public health and the clinical field in large parts of the world.

We found a large variety of instruments used, and the majority of the included studies (80 out of 120) presented a health literacy definition as part of the study's theoretical background. However, there seems to be an inconsistency between the definitions and the instruments in a significant number of the studies. In almost half of the studies, health literacy was defined in a broad perspective (including aspects such as social health literacy competence and the ability to process and appraise health information) while using instruments with a narrower focus (measuring basic skills and knowledge). As a result, there is a risk of missing important information about health literacy considered important in the studies' initial understanding of the concept. This concern was previously addressed by health literacy researchers.[23] Although numerous systematic reviews have reported on the diversity of understandings of health literacy and the various use of instruments not aligned to the definitions in current research,[13, 16, 19, 20-21] the current study is to our best knowledge the first review to investigate the relationship between health literacy definitions and instruments from the perspective of an established health literacy theory. We hope our work contributes to making explicit where the problem might be rooted and what strategy could be used to move forward to better align HL measurement with definitions of HL.

The use of instruments focusing on functional health literacy, such as Newest Vital Sign and Test of Functional Health Literacy, were predominant in our review. This finding corresponds with previous findings.[38, 20] The widespread use of these instruments is justified by the fact that they are screening tools—that is, they are quick, are available in an ever-increasing number of languages, and can be adapted to different settings.[38] Despite the increased development of questionnaires with a broader comprehensive focus, it might take more time for these instruments to become established in the health literacy research field. A broad range of skills and tasks covering functional, interactive, and critical domains needs to be included in measures of health literacy in order to capture the definitions.[39] In addition, most of the

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focus has been on measuring individual health literacy, but all sides of the health literacy equation (including health systems and health professionals) and social processes are important to research in order to get a comprehensive understanding of the phenomenon. Definitions should be the basis of measurement. However, measurement should continually inform definitions.[40] Our study highlights the discrepancy between definitions and measurements as well as the narrow focus of health literacy in a large amount of empirical research.

Nutbeam's description of three levels of health literacy inspired the categorization of definitions and instruments.[26] Initially, we planned to distinguish among the three levels. However, the line between Levels 2 and 3 was challenging to distinguish. Therefore, it was decided to merge the two latter categories. In a recent publication from 2020, Nutbeam provided a more thorough description of the three levels with a more detailed explanation of how the levels should be understood.[39] Perhaps, this would have contributed to a clearer guidance in our work with the categorization. However, this material was not available at the time of our work and, in general, this situation illustrates the challenge of adapting a theoretical model into practise.

The interpretation of terms used in definitions and instruments that guided the choice of category also represented some challenges. These were resolved through discussions both in pairs and as a research team. For instance, definitions describing health literacy using terms like "knowledge" and "beliefs" were included in Category 1 while definitions using terms such as "appraise" and "understand and process" were included in Category 2. An interesting finding is that, among the eight studies providing a health literacy definition from Category 1, the majority (n=5) were in the context of mental health. The predominant reference in these studies was the definition provided by Jorm and colleagues who, unlike others, defined health

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literacy as being linked to "beliefs" and "attitudes".[36, 37] Whether these terms are more closely linked to mental health literacy challenges compared to other more "physical" health literacy is not clear but would be interesting for further investigation.

Another aspect worth noting is that we discovered that many publications did not cite the primary source of the health literacy definitions, but instead referred to secondary sources (other researchers presenting definitions of health literacy). Referring to the original sources should be the first choice and would perhaps make it easier for readers to recognise the definition's affiliation.

The present study has some limitations. First, this study was designed to analyse and describe health literacy research in two randomly selected months. This period may not be representative of health literacy research in general. However, a large number of health literacy studies are published every year. A random selection can therefore give a good picture of health literacy research. To ensure that these months were not unique in terms of the number of articles published, we performed the same search strategy using the same databases for one other randomly selected month in 2019. This search yielded nearly the same number of articles. Second, we did not conduct a quality assessment screening of the included studies. This was considered less relevant for the current study as the aim of the current study was to explore connections between health literacy definitions and instruments rather than to assess methodology. Furthermore, searches were limited to English language only. It is possible that similar studies may have been published in languages other than English.

Conclusion

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There seems to be an inconsistency between the definitions and the instruments used in a significant part of current health literacy research. This situation raises the risk of missing important information about health literacy that was argued as important in studies' initial understanding of the concept. This gap should be taken into consideration in future health literacy research.

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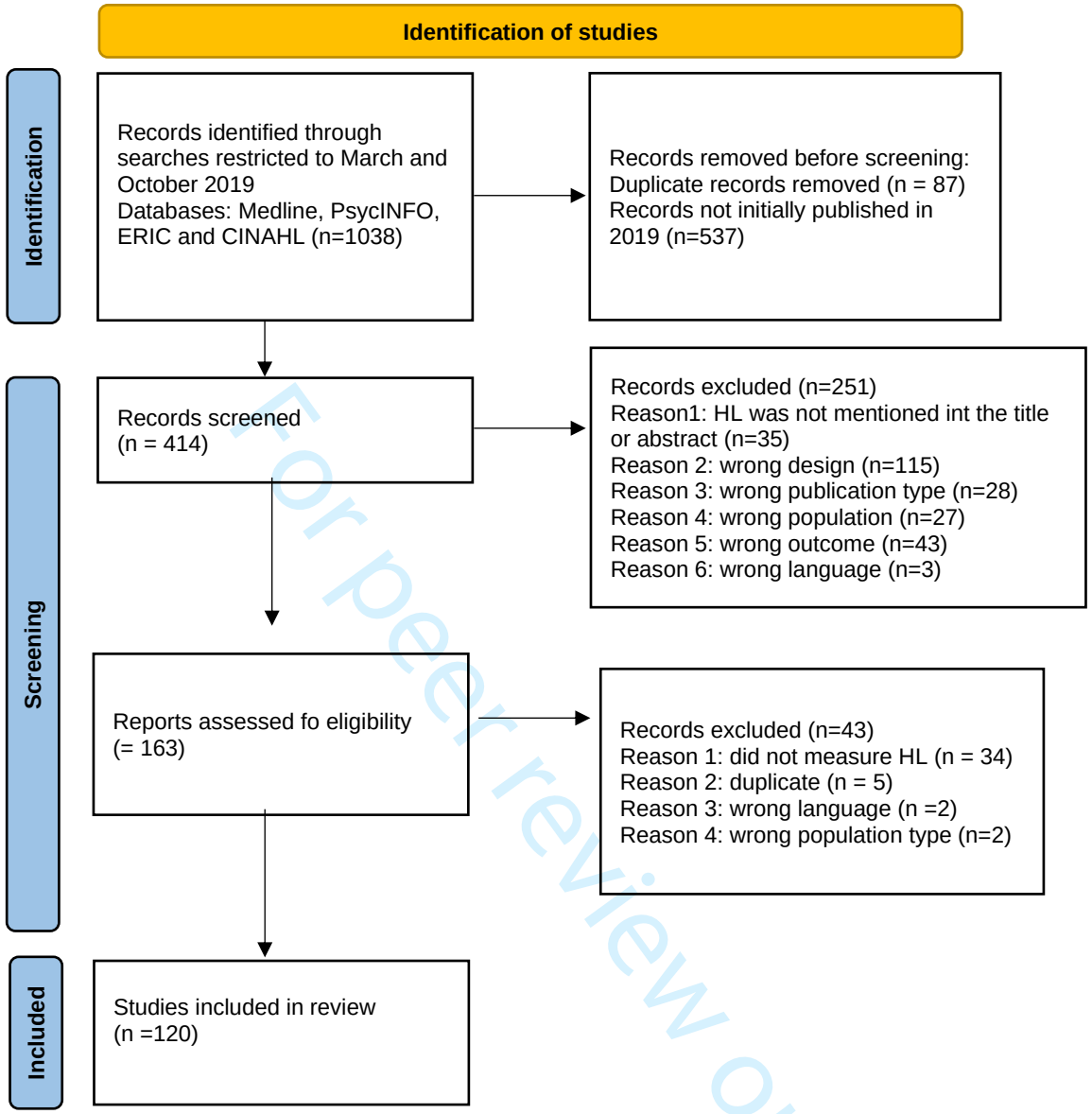
Contributors: KU, MA and AW initiated the project and KU led the project. KHU, MAH, AW, SH, CB and ML designed the study. KHU, MAH, AW, SH, CB and ML selected the articles, and extracted the data. KHU, MA and AW wrote and revised the manuscript. All authors contributed to drafting the manuscript and all listed authors meet authorship criteria.

Competing interest statement: No competing interests

Ethics statements: Patient consent for publication not required.

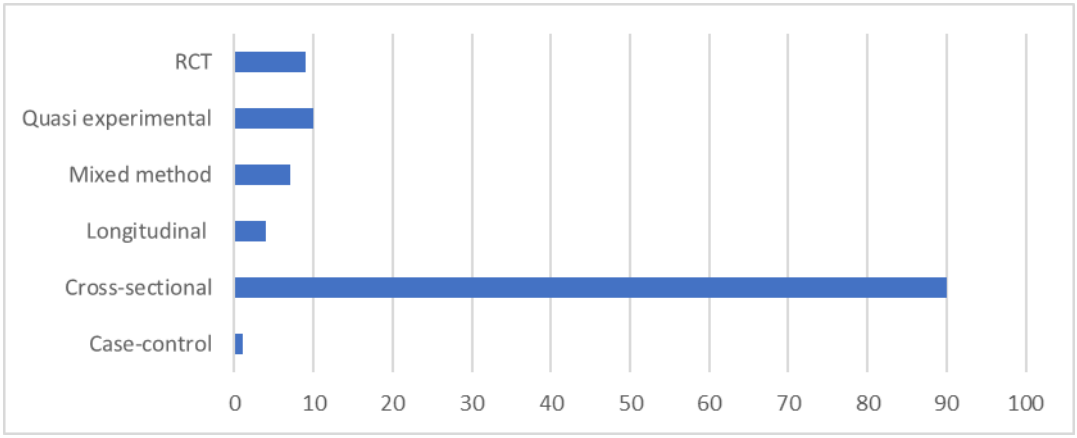
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Figure 1 Flow diagram shows the study selection process



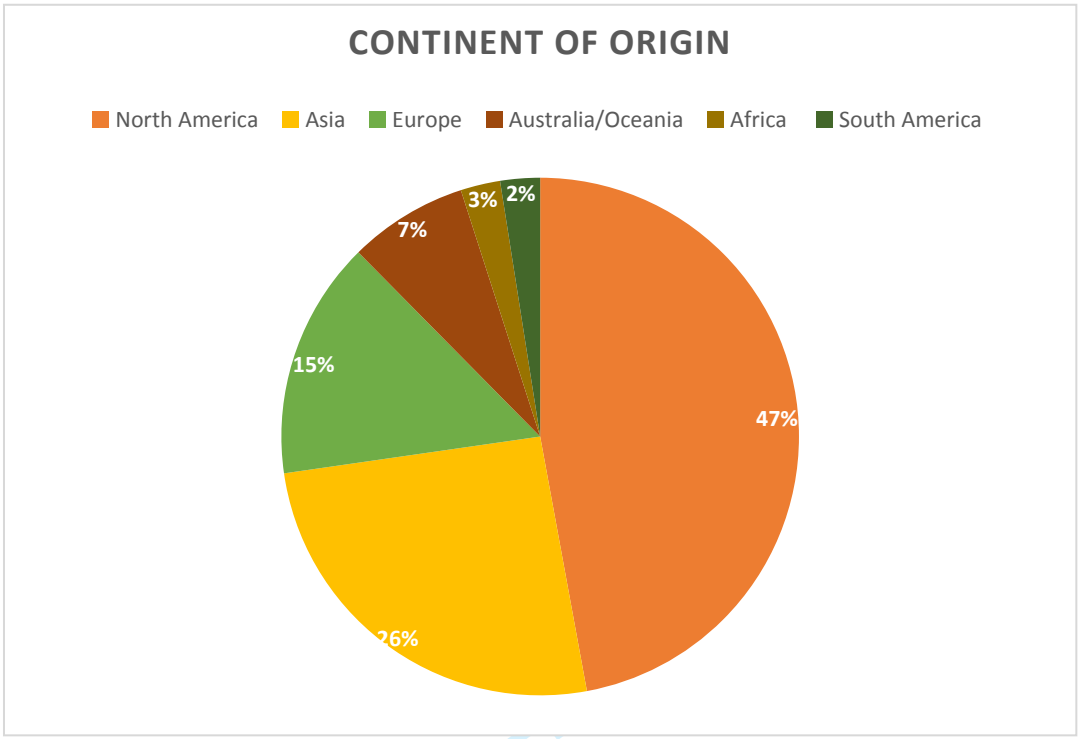
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Figure 2. Overview of study designs of included studies



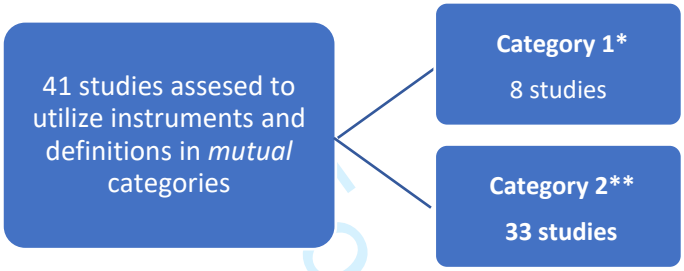
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Figure 3. Studies' continent of origin



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Figure 4. Studies categorized with the same level of HL definitions and HL instruments



**Category 1: basic reading and writing skills, disease-specific knowledge and practical skill competences*

***Category 2: including skills to communicate and interact with healthcare providers and the ability to interpret and critically analyse health information*

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Search history

Databases	Medline
Date	03.04.2020
Search history	1 Health Literacy/ (5393) 2 health literacy.ti,ab,kw. (8163) 3 1 or 2 (10010) 4 limit 3 to (dt=20191001-20191031 or rd=20191001-20191031) (324) 5 limit 3 to (dt=20190301-20190331 or rd=20190301-20190331) (210) 6 4 or 5 (532)
Number of hits	532
Comments	Limited to Entry Date (March + October 2019) from https://wkhealth.force.com/ovidsupport/s/article/Limit-by-date-range-in-Ovid

Databases	PsycINFO
Date	03.04.2020
Search history	1 exp health literacy/ (2991) 2 Health Literacy.ti,ab,id. (3744) 3 1 or 2 (4291) 4 limit 3 to up="20191001-20191031" (30) 5 limit 3 to up="20190301-20190331" (18) 6 4 or 5 (48)
Number of hits	48
Comments	Limited to Entry Date (March+ October 2019) from https://wkhealth.force.com/ovidsupport/s/article/Limit-by-date-range-in-Ovid

Databases	CINAHL with Full Text
Date	03.04.2020
Search history	S5 S3 AND S4 Search modes - Boolean/Phrase S4 EM 201903* or EM 201910* or RD 201903* or RD 201910* Search modes - Boolean/Phrase S3 S1 OR S2Search modes - Boolean/Phrase S2 "Health Literacy" Search modes - Boolean/Phrase S1 (MH "Health Literacy") Search modes - Boolean/Phrase
Number of hits	419

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Search history

Comments	Limited to Entry Date (March+ October 2019

Databases	ERIC
Date	03.04.2020
Search history	S2 "Health Literacy" Limiters - Date Published: 20190101-20191231 Search modes - Boolean/Phrase S1 "Health Literacy" Search modes - Boolean/Phrase
Number of hits	39
Comments	Searched for Publication Date - 2019 (All year)

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Appendix 2 Coding for the data extractions

1 .Does this study measure health literacy?	2. Perspective	3 .Design	4. Country of origin	5. Status population	6. Type of disease	7. Other disease
Yes No	Public health Clinical health care setting	RCT Quasi experimental Mixed method Longitudinal Cross-sectional Case-control		Healthy With a diagnose	Infectious- and parasite diseases Cancer Endocrine diseases Mental illness Diseases in the nervous system Diseases in the ear Diseases in the circulatory system Diseases in the respiratory system Diseases in the digestive system Diseases in the musculoskeletal system and connective tissue Diseases in the urine- and genital organs Pregnancy, birth, postnatal period	

8.Is HL defined?	9. Reference for HL defintion:	10. HL definition category
Yes No		Category 1 Category 2

11. HL instrument used	12. Reference for instrument	13. Provide reference	14. Type of HL instruemnt	15. HL instrument category
Name:	Yes No		Generic Disease specific	Category 1 Category 2

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 2, line 6
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 2, line 32
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 4-5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 5, line 44
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 6, line 56
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 6, line 23
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 7, line 3
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 7, line 3
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 6, line 56
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 6, line 56
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Not relevant
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Not relevant
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Not relevant
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Not relevant
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Not relevant
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Not relevant

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Not relevant
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Not relevant
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 8, line 38
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 9, line 12
Study characteristics	17	Cite each included study and present its characteristics.	Page 10, line 9
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Not relevant
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Not relevant
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Not relevant
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Not relevant
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Not relevant
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Not relevant
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Not relevant
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Not relevant
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 19, line 37
	23b	Discuss any limitations of the evidence included in the review.	Page 21, line 44
	23c	Discuss any limitations of the review processes used.	Page 21, line 44
	23d	Discuss implications of the results for practice, policy, and future research.	Page 22, line 24
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 3, line 23
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Page 3, line 23

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Page 7, line 55
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 24, line
Competing interests	26	Declare any competing interests of review authors.	Page 25, line 3
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Appendix 2

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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"Definitions and measurement of health literacy in health and medicine research: a systematic review"

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Secondary Subject Heading:	Health services research
Keywords:	Public health < INFECTIOUS DISEASES, GENERAL MEDICINE (see Internal Medicine), PUBLIC HEALTH

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"Definitions and measurement of health literacy in health and medicine research: a systematic review"

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Abstract

Objectives: The way health literacy is understood (conceptualized) should be closely linked to how it is measured (operationalized). This study aimed to gain insights into how health literacy is defined and measured in current health literacy research and examine the relationship between health literacy definitions and instruments.

Design: Systematic review in accordance with the PRISMA statement.

Data sources: Medline, PsycINFO, ERIC, and CINAHL databases were searched for articles published intwo randomly selected months (March and October) in 2019.

Eligibility criteria: We included articles with a quantitative design that measured health literacy, were peer-reviewed and original, was published in English language and with a study population over16 years.

Data extraction and synthesis: Six researchersscreened the articles for eligibility and extracted the data independently. All health literacy definitions and instruments were considered in relation to Category 1 (describing basic reading and writing skills, disease-specific knowledge, and practical skills) and Category 2 (social health literacy competence and the ability to interpret and critically assess health information). The categories were inspired by Nutbeam's descriptions of the different health literacy levels.

Results: 120 articles were included in the review, 60 within public health and 60 within clinical health. The majority of the articles (n=77) utilized instruments from Category 1. In total, 79 of the studies provided a health literacy definition; of these, 71were in Category 2 and 8 in Category 1. In almost half of the studies (n=38), health literacy was defined in a broad perspective (Category 2), but measured in a more narrow focus (Category 1)

Conclusion: Due to the high degree of inconsistency between health literacy definitions and instruments in current health literacy research, there is a risk of missing important information

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about health literacy that was argued to be important in studies' initial understanding of the concept.

PROSPERO registration number: CRD42020179699

Strengths and Limitations of This Study

- To our knowledge, this is the first systematic review to investigate connections between health literacy definitions and instruments used in current health literacy research
- The health literacy definitions and instruments were categorized based on a health literacy model described by Nutbeam
- The initial plan of assessing health literacy definitions and instruments for three categories was changed to two due to difficulties in distinguishing between two of the categories
- Searches were limited to two randomly selected months

Word count: 5351

Introduction

Health literacy is usually understood as cognitive and social skills which determine the motivation and ability to understand and use health information,[1] and adequate health literacy is seen as a prerequisite for healthy behaviours. Researchers have increasingly worked to identify challenges associated with health literacy and investigate the role they play in an individual's ability to comprehend self-care information and its relationship to health outcomes.[2] Empirical studies have reported that low health literacy is associated with poor health-related outcomes, such as high hospital admission rates,[3-4] low participation in

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preventive activities,[5-9] poor self-management of chronic conditions,[9] poor disease outcomes, and high mortality.[11-13]

The concept of health literacy emerged in the 1970s when health education was viewed as social policy.[14] A topic-specific query in the PubMed tools reveals an exponential growth of articles about health literacy, with 129 references between 1986 and 1990 compared to more than 8000 in the last five years. Today, health literacy is seen as a global goal to enhance health promotion through improved education and communication strategies to improve health outcomes.[15]

Health literacy is defined in numerous ways.[14, 16-18] In a systematic review by Sorensen et al.,[14] 17 different definitions and 12 conceptual models were reported on health literacy. Another systematic review found that 34 research articles between 2010 and 2015 had an explicit objective to define the concept of health literacy.[17] Moreover, the literature has reported the use and development of more than 150 health literacy instruments over the last decade.[19-21] Traditionally, health literacy approaches have focused on individual skill deficits and health education based on the communication of factual information on health risks as well as how to use the health system[22] while the majority of health literacy research has used instruments measuring reading and numeric skills.[23] However, in recent years, more multi-dimensional perspectives and instruments measuring health literacy have been introduced, such as the Health Literacy Questionnaire[24] and the Health Literacy Survey European Questionnaire 47.[25]

The current and sometimes confusing use of various and inconsistent interpretations of health literacy is a challenge for the development of valid and reliable measurements.[16] In 2000, Nutbeam proposed a health literacy model that is now widely cited in the health literacy literature and is by many health literacy researchers seen as useful in analysing health literacy

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abilities required in various health situations. [26] According to Nutbeam, health literacy contains three different levels, progressing from basic skills in reading and writing (functional health literacy) to the ability to derive meaning from different forms of communication and apply new information to changing situations (interactive health literacy) and to more advanced cognitive skills which, together with social skills, can be applied to critically analyse information the ability to achieve policy and organizational changes (critical health literacy).

Different understandings of health literacy and different measurement tools may be useful as they complement each other and provide different perspectives. However, the way health literacy is understood (conceptualized) should be closely linked to how it is measured (operationalized) in each study context.[23] Ngyen et al. has described this as a "conceptual stumbling block" that needs to be resolved for the field to progress.[23] A first useful step for addressing this might be to systematically explore how it appears in current health literacy research. Hence, by performing a systematic review, our aim is to gain insights into how health literacy is defined and measured in current health literacy research. In particular, we will examine the relationship between health literacy definitions and instruments. This review may increase our understanding of potential conceptual and methodological challenges or gaps that need to be addressed in future research.

Methods

This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement[26] and registered in PROSPERO (No. CRD42020179699): https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=179699. The review was designed with a time frame limited to two randomly selected months in 2019 (March and

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October). Due to a high number of health literacy articles published every year, random selection was chosen in order to reflect current health literacy research. To ensure that March and October were not unique in terms of the number of articles published, we performed the same search strategy using the same databases for one other randomly selected month in 2019. This search yielded nearly the same number of articles.

Search Strategy

Systematic literature searches were conducted in collaboration with a trained librarian (3.4.2020). The Medline, PsycINFO, ERIC, and CINAHL databases were searched for the term "health literacy". For all databases except for ERIC (where this was not possible), the search was automatically restricted to two randomly selected months in 2019: March and October. Citations in ERIC were manually assessed for articles published in March 2019 and October 2019.

The search yielded a total number of 1038 citations. Endnote™ Version X9 was used to manage the generated research articles. After removing duplicates, 951 citations remained. All records not published for the first time in 2019 were removed, leaving 414 articles for screening. (See Appendix 1 for the search history).

Selection Criteria

The inclusion and exclusion criteria were developed a priori. The 414 published articles were distributed among six researchers (KHU, AKW, MHA, CB, SH, MHL), who worked in pairs. The articles were included if they fulfilled the following inclusion criteria: 1) "health literacy"

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was mentioned in the title or abstract, 2) the article was peer reviewed, 3) the research was original, 4) it used a quantitative design, 5) it was published in the English language, and 6) the study population included individuals over 16 years of age.

To reduce selection bias, the researchers screened the articles for eligibility according to the criteria independently. Then they met in pairs to compare their results, resolve any conflicting opinions, and decide whether to include each article. Conflicting opinions in pairs were presented and solved by the whole team.

Data Extraction (Selection and Coding)

To achieve consistency in the data extracted from the included articles, an Excel spreadsheet was created. The initial question in this spreadsheet was whether health literacy was measured. If the answer was no, the article was excluded. The data extracted from the articles included information about the study design and context, such as country of origin and whether the study was conducted in a public health or in a clinical health setting. For clinical studies that included participants with health problems, the different types of diseases were categorized according to the International Statistical Classification of Diseases and Related Health Problems (ICD, WHO, 2010).[27] For the public health studies, we categorized type of study populations (Table 1). Furthermore, data included information about whether the instruments were generic or disease specific the. Also, we extracted whether it was used in combination with other health literacy instruments, and whether a reference was provided for the instrument.

The published articles were analysed regarding which health literacy definitions and instruments were used. Before the data extraction, a pilot assessment of three studies was performed to determine the feasibility of the data extraction tool. The initial plan was to

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assess health literacy definitions and instruments for three categories 1) functional, 2) interactive, and 3) critical health literacy, inspired by Nutbeam' s description of different health literacy levels [22]. As a result of the pilot testing, the number of categories was changed due to difficulties in distinguishing between the two latter levels. All health literacy definitions and health literacy instruments were therefore considered in relation to two categories; Category 1 included definitions and instruments describing basic reading and writing skills, disease-specific knowledge, and practical skill competencies needed to function in everyday situations and Category 2 included definitions and instruments that also described health literacy as skills to communicate and interact with healthcare providers as well as the ability to interpret and critically analyse health information. (Appendix 2 illustrates the coding for the data extractions).

The researchers first extracted data and considered definitions and instruments related to the two categories independently. They then met in pairs to compare their results and resolve any conflicting opinions. The remaining conflicting opinions within pairs were presented and solved by the whole team in a group meeting.

Patient and public involvement

No patient involved

Results

The search yielded 1038 articles in total (see the Appendix for the search history). After removing duplicates, 951 articles remained. Of the 163 articles that were read in full texts, one was a duplicate, and 39 were excluded because they did not measure health literacy. Ultimately, 120 articles were included (see Figure 1).

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Figure 1 about here

Characteristics of Included Studies

Sixty articles described studies with a clinical health focus, and 60 were conducted in public health settings. The clinical studies included a great variety of diseases, but the most frequent diseases were those in the circulatory system (n=11) and endocrine diseases (n=9) followed by mental illness (n=6) and cancer diseases (n=6). In the public health studies, the most frequent study group was the general population (n=19), looking into aspects such as vaccination programs, oral care and mental health. Furthermore, a major part of the public health studies included students (n=13) mainly represented by health professional students. A number of 6 studies focused on parents/care givers, while 5 studies focused on the elderly population.

Table 1. Overview of study populations in clinical health studies and in the public health studies

Clinical health studies		**Public health studies**	
Populations classified by Diseases and Related Health Problems	n	*Type of study population*	n
Diseases in the circulatory system	11	General population	19
Endocrine diseases	9	Students	13
Mental illness	6	Parents/care givers	6
Cancer	6	Elderly people	5
Diseases in the respiratory system	4	Immigrants	3
Diseases in the urine- and genital organs	4	Work related populations	3
Infectious- and parasite diseases	3	Young adults	2
Diseases in the ear	2	Veterans	2
Diseases in the musculoskeletal system and connective tissue	2	Health personell	2
Diseases in the nervous system	1	Men	1
Diseases in the digestive system	1	Women	1

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		Pet owners	1
Pregnancy, birth, postnatal period	1	Relatives of patients with cancer	1
Other: General chronic conditions (n=2), chronic pain (n=3) patients in primary care (n=2) next of kind (n=2), patients in specialist care (n=1)	10	Churchgoers	1
Total	60	**Total**	60

The majority of the studies utilized a cross-sectional design (n=90). Nine studies utilized a randomized controlled design, and 10 studies were quasi-experimental. Four of the studies used a longitudinal design while one was a case-control study. Figure 2 presents an overview of all study designs.

Figure 2 about here

As Figure 3 indicates, the majority of studies were conducted in North America (n=56) and Asia (n=31). The fewest studies were performed in Africa (n=4) and South America (n=3).

Figure 3 about here

Health Literacy Definitions and Instruments

Out of the 120 included studies, 88 utilized generic health literacy instruments while 32 were context-specific. Eleven studies used a combincation of health literacy instruments. A number of 77 studies utilized instruments from Category 1 (describing basic reading and writing skills, disease-specific knowledge, and practical skills), whereas 43 instruments were from

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Category 2 (described communication and interaction skills and/or the ability to interpret and critically analyse health information). The most frequently used HL instruments in Category 1 were Newest Vital Sign[28] (n=19) and Test of Functional Health Literacy [29] (n=13). The most frequently used HL instruments in Category 2 were The European Health Literacy Survey Questionnaire[30] and The eHealth Literacy Scale[31] (n=10).

Of the 120 articles, 79 provided a health literacy definition. Of these, 46 were public health studies and 33 clinical health studies. Only eight studies provided a definition addressed to Category 1. Five out of these eight studies focused on mental health challenges.

The most frequently used reference in Category 2 was Nutbeam.[22,32,33] The most frequently used reference in Category 1 definitions was a definition provided by Jorm and colleagues.[33-35] Table 2 gives an overview of the text in the most cited health literacy definitions. Words in italics are examples of key words considered important in the decision to adding the definition to Category 1 or Category 2. For instance, definitions that described health literacy with terms such as "knowledge", "skills" and "attitude" were referred to Category 1, while definitions with terms like "appraise" and "social skills" were referred to Category 2.

Table 2. The most frequently used references for health literacy definitions in the included studies

Reference	**Definition**	**Number of studies**
Category 1		
Jorm (1997) Jorm (2012)	"knowledge about appropriate treatment options; and *attitudes* that facilitate recognition and treatment-seeking" "*knowledge and beliefs* about mental disorders which aid their recognition, management or prevention"	5
American Medical	"The *constellation of skills, including the ability* to perform basic reading and numerical tasks required to function in the healthcare environment"	1

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Association (AMA) (1999)		
Category 2		
Nutbeam (1998, 2000, 2015)	"The *cognitive and social skills* which determine the motivation and ability of individuals to gain access to understand and use information in ways which promote and maintain good health."	19
Sørensen (2012)	"Health literacy entails people's knowledge, motivation and competences to access, *understand*, *appraise*, and *apply* health information in order to make judgments and take decisions in everyday life concerning healthcare, disease prevention and health promotion to maintain or improve quality of life during the life course."	14
The Institute of Medicine (2004)	"The individuals' capacity to *obtain, process* and understand basic health information and services needed to make appropriate health decisions."	11

Regarding the connection between health literacy definitions and instruments in the 79 relevant studies, 41 articles used health literacy definitions and instruments characterized at the same level. Thirty-three of these were in Category 2, and 8 were in Category 1 (see Figure 4 and Table 3)

Figure 4 about here

In the remaining 38 studies, there was a disconnect between levels of health literacy definitions and instruments. In all of these, health literacy definitions were from Category 2, and all instruments were in Category 1 (see Table 4).

Table 3. Studies categorized with mutual levels of HL definitions and HL instruments (n=41)

Study reference	**Study context**	**Design**	**Health literacy definition reference**	**Cat.**	**Health literacy instrument**	**Cat.**
Category 1						

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Clough et al., 2019	Public health	RCT	Jorm et al., 1997	1	Mental Health Literacy Scale (O'Connor and Casey, 2015)	1
Clough et al., 2019	Public health	Cross-sectional	Jorm et al. 1997	1	Mental Health Literacy Scale (O'Connor & Casey, 2015)	1
Lichtveld et al, 2019	Public health	Cross-sectional	Finn and O'Fallon, 2019 Gray, 2018	1	Developing and validating a questionnaire measuring environmental health literacy: Measuring media-specific knowledge, attitudes, and behaviors	1
Nye et al., 2019	Public health	Longitudinal	U.S. Department of Health and Human Services, 2010	1	The Upper Peninsula Oral Health Assessment Survey	1
Williston et al., 2020	Public health	Longitudinal	Jorm et al., 1997	1	Mental Health Literacy Scale (O'Connor and Casey, 2015)	1
Cehuen Neto et al., 2019	Clinical	Cross-sectional	American Medical Association (AMA) ad hoc committee on health, 1999	1	Short Test of Functional Health Literacy in Adults (Parker, 1995)	1
Joncho et al., 2019	Clinical	Cross-sectional	Jorm et al., 1997	1	Developed within the study context: Mental Health Literacy Questionnaire	1
Waldmann et al.,2020	Clinical	Cross-sectional	Jorm et al., 2012	1	Mental Health Knowledge Schedule. Depression Literacy Scale (Wilson, 2015)	1
Category 2						
An et al., 2019	Public health	Cross-sectional	Nutbeam, 2000	2	The Health Literacy Scale (Chew, 2004)	2
Azizi et al., 2019	Public health	Mixed method	Nutbeam, 2000	2	Health Literacy Scale for Workers (HELSW)	2
Brandsetter et al., 2019	Public health	Cross-sectional	Sørensen, 2012	2	The European Health Literacy Survey (HLS-EU) (Sørensen et al., 2013)	2
Evans et al., 2019	Public health	Cross-sectional	Sørenesen, 2012	2	European Health Literacy Survey Questionnaire (Sørensen et al., 2013)	2
Eo et al., 2019	Public health	Cross-sectional	Nutbeam, 2000	2	The Health Literacy Assessment Scale for Asian Immigrant Women (And and Yang, 2015)	2
Erunal et al., 2019	Public health	RCT	Nutbeam, 2000	2	Turkish Health Literacy Scale -32 (Okyay et al., 2016)	2
					European Health Literacy Survey Questionnaire -HLS	

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Fernandez Gutierrez et al., 2019	Public health	Quasi experimental	Nutbeam, 2000	2	EU-16 (Sørensen et al., 2013) Health Literacy App Questionnaire + five practical tests	2
Goto et al., 2019	Public health	Cross-sectional	Nutbeam, 1998	2	The Functional Communicative and Critical HL Tool (Ishikawa et al., 2008)	2
Gûner et al., 2019	Public health	Cross-sectional	Nutbeam, 2000	2	Questionnaire developed within the study context (consisting of 21 questions about HL knowledge, practice, and attitudes)	2
Hu et al., 2019	Public health	Cross-sectional	Nutbeam, 2000	2	Health Literacy Management Scale (Jordan, 2013)	2
Indino et al 2019	Public health	Cross-sectional	Nutbeam, 2015	2	The Functional Communicative and Critical Health Literacy Tool (Ishikawa et al., 2008)	2
Kaper et. al, 2019	Public health	Quasi experimental	Kwan, 2006	2	Questionnaire developed within the study context by combining subscales from other questionnaires	2
Kobayashi et al., 2019	Public health	Mixed method	Sørensen, 2012	2	European Health Literacy Survey Questionnaire (Sørensen et al., 2013)	2
Logullo et al., 2019	Public health	Cross-sectional	Sørensen, 2012	2	Short Assessment of Health Literacy for Portuguese Speaking Adults (SAHL-PA) (Apolonario et al., 2012)	2
Ma et al., 2019	Public health	Cross-sectional	Norman et al., 2006	2	The eHealth Literacy Scale (Norman, 2006)	2
Oh et al., 2019	Public health	Cross-sectional	World Health Organization, 2009	2	The eHealth Literacy Scale (Norman, 2006)	2
Oliffe et al, 2019	Public health	Cross-sectional	Peerson and Saunders, 2009, O'Brian et al., 2005	2	The Health Literacy Questionnaire (Osborn et al., 2013)	2
Pobhirun and Pinitsoontorn, 2019	Public health	Cross-sectional	Nutbeam, 1998	2	A questionnaire developed within the study context including 6 dimensions: cognitive skills, access, communication skills, self-management, media literacy and decision-making skills	2
Rababah et al, 2019	Public health	Cross-sectional	World Health Organization, 2013, Sørensen, 2012	2	The Health Literacy Questionnaire (Osborn et al., 2013)	2

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Ruëgg and Abel, 2019	Public health	Cross-sectional	Selden et al., 2000	2	Short survey tool for public health and health promotion research (Abel, 2014)	2
Solhjoo Niloofar et al., 2019	Public health	Mixed method	Sørensen, 2012	2	The eHealth Literacy Scale (Norman, 2006)	2
Uysal et al., 2019	Public health	Cross-sectional	Sørensen, 2012	2	European Health Literacy Survey Questionnaire (Sørensen et al., 2013)	2
Degan et al., 2019	Clinical	Cross-sectional	World Health Organization, 2009	2	The Health Literacy Questionnaire (Osborn et al., 2013)	2
Ernsting et al., 2019	Clinical	Cross-sectional	Sørenesen, 2012	2	European Health Literacy Survey Questionnaire - short form (Sørensen et al., 2013) and The eHealth Literacy Scale (Norman, 2006)	2
Guclu et al., 2019	Clinical	Cross-sectional	World Health Organization, 2009	2	European Health Literacy Survey Questionnaire (Sørensen et al., 2013)	2
Lim et al., 2019	Clinical	Cross-sectional	Osborne, 2013	2	Health Literacy Questionnaire (Osborne et al., 20139	2
Lu et al., 2020	Clinical	Cross-sectional	Sørensen et al., 2012	2	European Health Literacy Survey Questionnaire -HLS EU-16 (Sørensen et al., 2013)	2
Lu et al., 2019	Clinical	Cross-sectional	Sørensen, 2012	2	European Health Literacy Survey Questionnaire -HLS EU-16 (Sørensen et al., 2013)	2
Nokes et al, 2019	Clinical	Quasi experimental	Norman et al., 2006 and 2011	2	The eHealth Literacy Scale (Norman, 2006)	2
Strømer et al., 2019	Clinical	Cross-sectional	World Health Organization, 1998	2	The Health Literacy Questionnaire (Osborn et al., 2013)	2
Zhang et al., 2019	Clinical	Cross-sectional	Sørensen, 2012	2	Heart Failure-specific Health Literacy scale (HFHL) (Matsuka et al., 2018)	2
Wang et al., 2019	Clinical	Cross-sectional	Kutner et al., 2003	2	Chinese Health Literacy Scale for Diabetes (Leung, 2013)	2
Wang et al., 2019	Clinical	Cross-sectional	Nutbeam, 2009	2	Diabetes Health Literacy Scale (Lee, 2018)	2

Table. 4 Studies categorized with conflicting levels of HL definitions and HL instruments (n=38)

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Study reference	Study context	Design	Health literacy definition reference	Cat.	Health literacy instrument	Cat.
Avci et al.,2019	Public health	Cross-sectional	US Department of Health and Human Services, 2000	2	Newest Vital Sign (NVS) (Weiss et al., 2005) and Rapid Estimate of Adult Literacy in Medicine (Davis, 1993)	1
Bonaccorsi et al., 2019	Public health	Cross-sectional	Sørensen, 2012	2	Newest Vital Sign (Weiss et al., 2005)	1
Bonaccorsi et al., 2019 II	Public health	Cross-sectional	Sørensen, 2012	2	Newest Vital Sign (Weiss et al., 2005)	1
Carducci et al., 2019	Public health	Crosa-sectional	Ratzan and Parker, 2000 Nutbeam, 2000	2	Test of Functional Health Literacy in Adults (TOFHLA) (Parker, 1995)	1
Flynn et al., 2019	Public health	Cross-sectional	American Dental Association Council, 2015	2	Oral Health Literacy Adults Questionnaire (Sistani et al., 2014)	1
Gaikwad, 2019	Public health	Cross-secttional	Ratzan and Parker, 2000	2	Rapid Estimation of Adult Literacy in Dentistry 30-word version (Lee et al., 2007)	1
Gûner et al., 2019	Public health	Cross-sectional	Nutbeam, 2000	2	Developed within the context of the study	1
Han et al.,, 2019	Public health	Cross-sectional	Sørensen, 2012, Nutbeam, 2000	2	Assessment of Health Literacy in Cancer Screening (AHL-C) (Han et al., 2014)	1
Himes et al., 2019	Public health	Cross sectional	Institute of Medicine, US, Committee HL, 2004	2	Chew et al.'s (2004) set of brief questions	1
Irvin et al., 2019	Public health	Cross-sectional	Peters et al., 2012	2	The Water Environmental Literacy Level Scale (WELLS) (Irvin et al., 2019)	1
Kaur et al., 2019	Public health	RCT	Healthy People, 2010 (Oral health)	2	TS-REALD Two-Stage Rapid Estimate of Adult Literacy (Stucky et al., 2011)	1
Kim et al., 2019	Public health	Cross-sectional	Ratzan and Parker, 2000	2	Three questions, each addressing oral, listening, and written literacies (no reference)	1
Kim et al., 2019			Joint Committee on	2		1

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	Public health	Quasi-experimental	National Education Standards, 1995		Knowledge questions developed in the study context	
Kino et al., 2020	Public health	Cross-sectional	Inst of Medicine, 2004	2	Threee indicators of HL (Haun et al., 2014)	1
Lin et al., 2019	Public health	Quasi-experimental	Nutbeam, 2000	2	Mandarin Health Literacy Scale (Lee et al., 2012)	1
Mackert et al., 2019	Public health	Cross-sectional	Berkman et al., 2010	2	Newest Vital Sign (NVS) (Weiss et al., 2005)	1
Tucker et al., 2019	Public health	Quasi experimental	Ratzan and Parker, 2000	2	Newest Vital Sign (NVS) (Weiss et al., 2005)	1
Van Wormer et al., 2019	Public health	Cross-sectional	Institute of Medicine, 2004	2	Oral Health Literacy in Adults Questionnaire (Sistani et al., 2013)	1
Weaver et al., 2019	Public health	Cross sectional	Ratzan and Parker, 2000 Nutbeam, 2000	2	Rapid Estimate of Adult Literacy (REALM) (Davis, 1993)	1
Anderson et al., 2019	Clinical	Cross-sectional	Defined within the study context (no reference provided)	2	Short Test of Functional Health Literacy in Adults (S-TOFHLA) (Parker, 1995)	1
Chen et al., 2019	Clinical	Cross-sectional	Nutbeam, 2000	2	Health Literacy Scale for Diabetes (Lee et al., 2016)	1
De Melo et al., 2019	Clinical	Cross-sectional	Cavanaugh, 2011	2	Short Test of Functional Health Literacy in Adults (Parker, 1995)	1
Kim et al, 2019	Clinical	RCT	Institute of Medicine, US, Committee HL, 2004	2	Short Test of Functional Health Literacy in Adults (Parker, 1995)	1
Lindahl et al., 2020	Clinical	Cross-sectional	Nutbeam and Kickbusch, 2000	2	The three-item Brief Health Literacy Screen (BHLS) (Cavanaugh et al., 2015)	1
Mayer et al., 2019	Clinical	Cross-sectional	Nutbeam, 1998	2	Developed within the context of the study	1
Meyers et al., 2019	Clinical	Cross-sectional	Institute of Medicine, 2004	2	Newest Vital Sign (Weiss et al., 2005)	1
Miranda, 2019	Clinical	Cross-sectional	Sørensen et al., 2012	2	Rapid Estimate of Adult Literacy Dutch version (REALM-D) (Davis, 1993)	1

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Mora-Pinzon et al., 2019	Clinical	Cross-sectional	Institute of Medicine, 2004	2	Short Test of Functional Health Literacy in Adults (S-TOFHLA) (Parker, 1995) Rapid Estimate of Adult Literacy (REALM-D) (Davis, 1993	1
Mock et al., 2019	Clinical	Cross-sectional	Institution of Medicine, 2004	2	Single Item Literacy Screening (SILS) (Morris, 2006), Short Test of Functional Health Literacy in Adults (S-TOFHLA) (Parker, 1995)	1
Noback et al., 2019	Clinical	Cross-sectional	Paasche-Orlow et al., 2007, Berkman et al., 2011, Pendlimari et al., 2012	2	Newest Vital Sign (NVS) (Weiss et al., 2005) Musculoskeletal specific literacy survey (LiMP questionnaire) developed within the study context	1
O'Conor et al., 2019	Clincial	Cross-sectional	Institute of Medicine Health Literacy, 2004	2	Short Test of Functional Health Literacy in Adults (S-TOFHLA) (Parker, 1995)	1
Penaloza et al, 2019	Clinical	Cross-sectional	Nutbeam, 1998 and Institute of Medicine, 2004	2	Short Assessment of Health Literacy Spanish (SAHL-S) (Lee et al., 2006)	1
Rafferty et al., 2019	Clinical	Cross-sectional	Institute of Medicine, 2004 and Sørensen, 2012	2	Questions focusing on health advice and information-seeking skills, oral literacy, and print literacy	1
Sany et al., 2019	Clinical	Quasi experimental	United States Health Resources and Services Administration	2	Test of Functional Health Literacy in Adults (TOFHLA) (Parker, 1995)	1
Schrivner et al., 2019	Clinical	Cross-sectional	Nutbeam, 2008	2	Three questions assessing health literacy (no reference)	1
Zhong et al., 2020	Clincial	Cross-sectional	Kindig, 2004	2	Newest Vital Sign (NVS) (Weiss et al., 2005)	1
Winokur et al., 2019	Clincial	Cross-sectional	Morrison et al., 2013	2	Newest Vital Sign (NVS) (Weiss et al., 2005)	1

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Williams et al, 2019	Clinical	Cross-sectional	Institute of Medicine, 2004	2	Newest Vital Sign (NVS) (Weiss et al., 2005)	1

Discussion

This systematic review aimed to gain insights into how current research defines and measures health literacy and, in particular, whether studies consistently used definitions and instruments. The high number of articles published in the defined timeframe shows that health literacy is of high research interest in both public health and the clinical field in large parts of the world.

We found a large variety of instruments used, and the majority of the included studies (79 out of 120) presented a health literacy definition as part of the study's theoretical background. However, there seems to be an inconsistency between the definitions and the instruments in a significant number of the studies. In almost half of the studies, health literacy was defined in a broad perspective (including aspects such as social health literacy competence and the ability to process and appraise health information) while using instruments with a more narrow focus (measuring basic skills and knowledge). As a result, almost half of the articles in our review lacked data on the participants` ability to critically appraise health information and their social health literacy competence despite the fact that the authors had stated such aspects to be health literacy. This concern has previously been addressed. Numerous systematic reviews have reported on the diversity of understandings of health literacy and the various use of instruments not aligned to the definitions in current research.[16-21] However, the current study is to our best knowledge, the first review to investigate the relationship between health literacy definitions and instruments systematically from the perspective of a health literacy model.

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The use of instruments that focused on functional health literacy (addressed to Category 1), such as Newest Vital Sign and Test of Functional Health Literacy, were predominant in our review.[28,29.] This finding corresponds with previous findings.[20,30] The widespread use of Newest Vital Sign and Test of Functional Health Literacy is justified by the fact that they are screening tools—that is, they are quick, are available in an ever-increasing number of languages, and can be adapted to different settings.[36] However, a broad range of skills and tasks covering functional, interactive, and critical domains needs to be included in measures of health literacy in order to capture the health literacy definitions used.[32] In recent years an increasing amount of multi-dimensional instruments have been developed. In our review, The European Health Literacy Survey Questionnaire and The eHealth Literacy Scale, both addressed for Category 2, were the multi-dimensional instruments most frequently used. [30,31] In light of recent years increased focus on a more multi-dimensional perspective of health literacy, and the fact that the current review only included studies from 2019, one could expect a higher user of multidimensional instruments. Instead, they represented only 43 instruments out of 120.

Clearly, our study highlights the discrepancy between definitions and measurements as well as the narrow focus of health literacy in a large amount of empirical research. An important question to raise is therefore how the findings from this specific study can guide future research strategies to overcome the identified inconsistency. In other words, as most health literacy researchers seem to base their research on a broad understanding of health literacy, what can be done to facilitate increased use also of broad measurements? A first step should be to make researchers aware about the existing mismatch in current research. Furthermore, it seems necessary to develop more instruments that can answer the research questions posed. Despite the high number of instruments, there still seems to be a need for questionnaires in the

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field of health literacy that captures more multidimensional dimensions of health literacy besides the functional aspects.

Nutbeam' s description of three levels of health literacy inspired the categorization of definitions and instruments.[22] Initially, we planned to distinguish among the three levels. However, the line between Levels 2 and 3 was challenging to distinguish. Therefore, it was decided to merge the two latter categories. In a recent publication from 2020, Nutbeam provided a more thorough description of the three levels with a more detailed explanation of how the levels should be understood.[37] Perhaps, this would have contributed to a clearer guidance in our work with the categorization. However, this material was not available at the time of our work and, in general, this situation illustrates the challenge of adapting a theoretical model into practise.

The interpretation of terms used in definitions and instruments that guided the choice of category also represented some challenges. These were resolved through discussions both in pairs and as a research team. For instance, definitions describing health literacy using terms like "knowledge" and "beliefs" were included in Category 1 while definitions using terms such as "appraise" and "understand and process" were included in Category 2. An interesting finding is that, among the eight studies providing a health literacy definition from Category 1, the majority (n=5) were in the context of mental health. The predominant reference in these studies was the definition provided by Jorm and colleagues who, unlike others, defined health literacy as being linked to "beliefs" and "attitudes".[34,,35] Whether these terms are more closely linked to mental health literacy challenges compared to other more "physical" health literacy is not clear but would be interesting for further investigation.

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Another aspect worth noting is that we discovered that many publications did not cite the primary source of the health literacy definitions, but instead referred to secondary sources (other researchers presenting definitions of health literacy). Referring to the original sources should be the first choice and would perhaps make it easier for readers to recognise the definition's affiliation.

The present study has some limitations. First, this study was designed to analyse and describe health literacy research in two randomly selected months. This period may not be representative of health literacy research in general. However, a large number of health literacy studies are published every year. A random selection can therefore give a good picture of health literacy research. Second, we did not conduct a quality assessment screening of the included studies. This was considered less relevant for the current study as the aim of the current study was to explore connections between health literacy definitions and instruments rather than to assess methodology. Furthermore, searches were limited to English language only. It is possible that similar studies may have been published in languages other than English.

The current review only included only quantitative measurements. However, qualitative approaches might provide valuable and more in-depth insights into the field. For future research, it would be interesting to also explore how qualitative research links health literacy definitions to research questions posed.

Conclusion

There seems to be an inconsistency between the definitions and the instruments used in a significant part of current health literacy research. This situation raises the risk of missing

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important information about health literacy that was argued as important in studies' initial understanding of the concept. This gap should be taken into consideration in future health literacy research. We hope our work contributes to making explicit where the problem might be rooted and that it can be useful in the discussion about strategies for moving forward to better align HL measurement with definitions of HL

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Competing interest statement: No competing interests

Ethics statements: Patient consent for publication not required.

Data availabillity: The data from the study will be made available upon request to the corresponding author.

Figure legends:

Figure 1 Flow diagram shows the study selection process

Figure 2 Overview of study designs of included studies

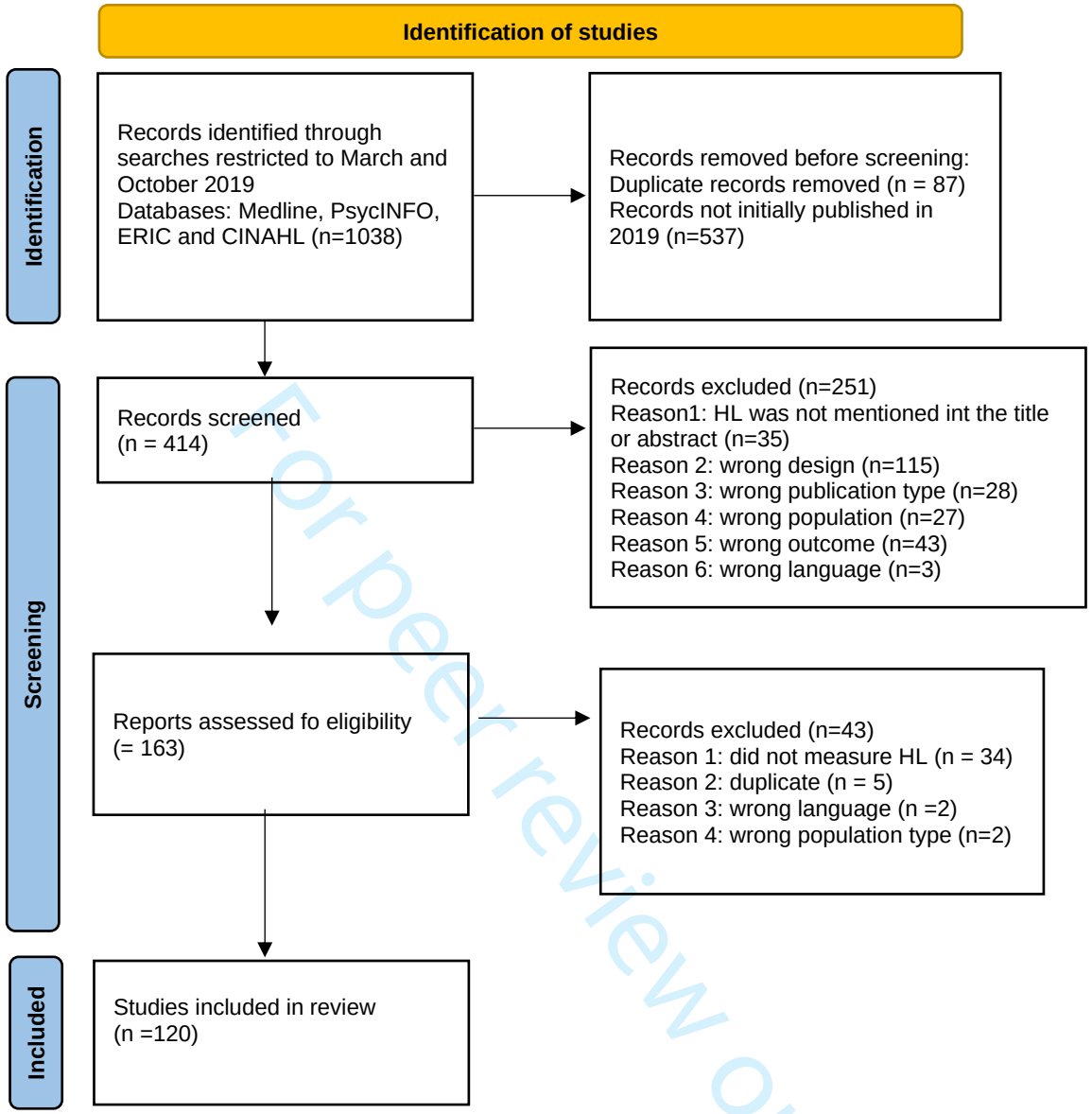
Figure 3 Studies' continent of origin

Figure 4 Studies categorized with the same level of HL definitions and HL instruments

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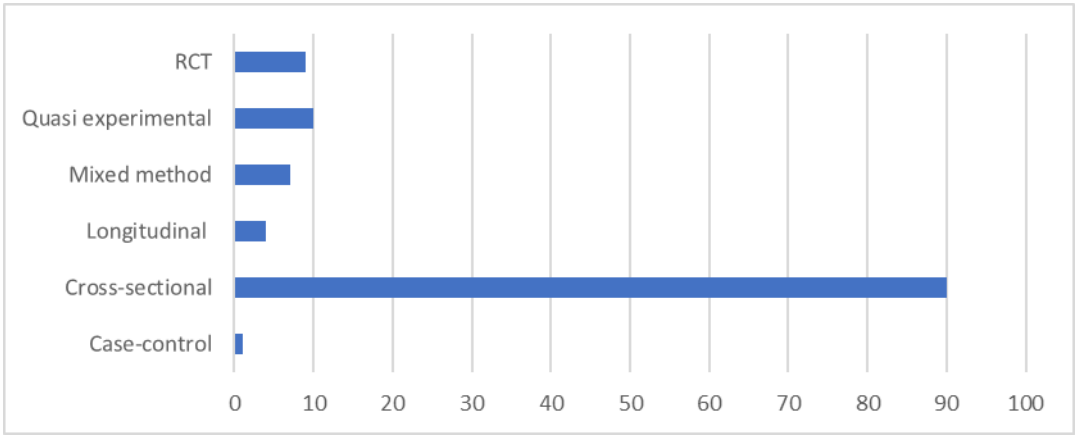
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Figure 1 Flow diagram shows the study selection process



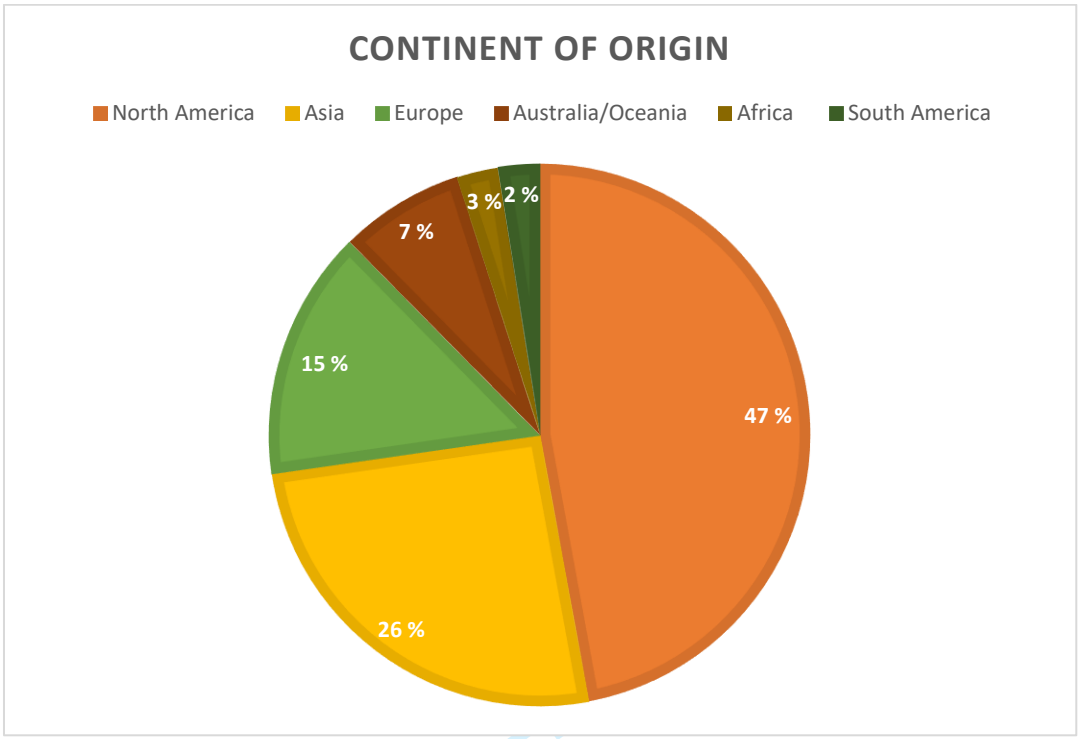
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Figure 2. Overview of study designs of included studies



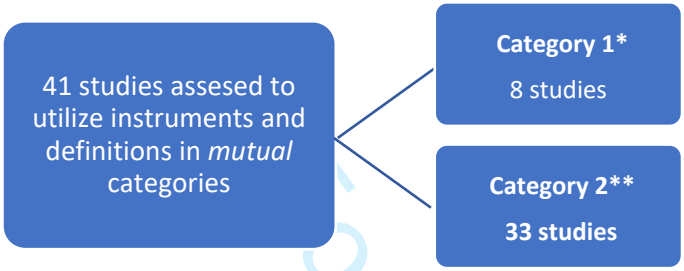
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Figure 3. Studies' continent of origin



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Figure 4. Studies categorized with the same level of HL definitions and HL instruments



**Category 1: basic reading and writing skills, disease-specific knowledge and practical skill competences*

***Category 2: including skills to communicate and interact with healthcare providers and the ability to interpret and critically analyse health information*

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Search history

Databases	Medline
Date	03.04.2020
Search history	1 Health Literacy/ (5393) 2 health literacy.ti,ab,kw. (8163) 3 1 or 2 (10010) 4 limit 3 to (dt=20191001-20191031 or rd=20191001-20191031) (324) 5 limit 3 to (dt=20190301-20190331 or rd=20190301-20190331) (210) 6 4 or 5 (532)
Number of hits	532
Comments	Limited to Entry Date (March + October 2019) from https://wkhealth.force.com/ovidsupport/s/article/Limit-by-date-range-in-Ovid

Databases	PsycINFO
Date	03.04.2020
Search history	1 exp health literacy/ (2991) 2 Health Literacy.ti,ab,id. (3744) 3 1 or 2 (4291) 4 limit 3 to up="20191001-20191031" (30) 5 limit 3 to up="20190301-20190331" (18) 6 4 or 5 (48)
Number of hits	48
Comments	Limited to Entry Date (March+ October 2019) from https://wkhealth.force.com/ovidsupport/s/article/Limit-by-date-range-in-Ovid

Databases	CINAHL with Full Text
Date	03.04.2020
Search history	S5 S3 AND S4 Search modes - Boolean/Phrase S4 EM 201903* or EM 201910* or RD 201903* or RD 201910* Search modes - Boolean/Phrase S3 S1 OR S2Search modes - Boolean/Phrase S2 "Health Literacy" Search modes - Boolean/Phrase S1 (MH "Health Literacy") Search modes - Boolean/Phrase
Number of hits	419

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Search history

Comments	Limited to Entry Date (March+ October 2019

Databases	ERIC
Date	03.04.2020
Search history	S2 "Health Literacy" Limiters - Date Published: 20190101-20191231 Search modes - Boolean/Phrase S1 "Health Literacy" Search modes - Boolean/Phrase
Number of hits	39
Comments	Searched for Publication Date - 2019 (All year)

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Appendix 2 Coding for the data extractions

1 .Does this study measure health literacy?	2. Perspective	3 .Design	4. Country of origin	5. Status population	6. Type of disease	7. Other disease
Yes No	Public health Clinical health care setting	RCT Quasi experimental Mixed method Longitudinal Cross-sectional Case-control		Healthy With a diagnose	Infectious- and parasite diseases Cancer Endocrine diseases Mental illness Diseases in the nervous system Diseases in the ear Diseases in the circulatory system Diseases in the respiratory system Diseases in the digestive system Diseases in the musculoskeletal system and connective tissue Diseases in the urine- and genital organs Pregnancy, birth, postnatal period	

8.Is HL defined?	9. Reference for HL defintion:	10. HL definition category
Yes No		Category 1 Category 2

11. HL instrument used	12. Reference for instrument	13. Provide reference	14. Type of HL instruemnt	15. HL instrument category
Name:	Yes No		Generic Disease specific	Category 1 Category 2

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 2, line 6
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 2, line 32
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 4-5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 5, line 44
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 6, line 56
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 6, line 23
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 7, line 3
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 7, line 3
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 6, line 56
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 6, line 56
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Not relevant
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Not relevant
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Not relevant
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Not relevant
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Not relevant
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Not relevant

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Not relevant
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Not relevant
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 8, line 38
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 9, line 12
Study characteristics	17	Cite each included study and present its characteristics.	Page 10, line 9
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Not relevant
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Not relevant
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Not relevant
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Not relevant
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Not relevant
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Not relevant
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Not relevant
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Not relevant
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 19, line 37
	23b	Discuss any limitations of the evidence included in the review.	Page 21, line 44
	23c	Discuss any limitations of the review processes used.	Page 21, line 44
	23d	Discuss implications of the results for practice, policy, and future research.	Page 22, line 24
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 3, line 23
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Page 3, line 23

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Page 7, line 55
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 24, line
Competing interests	26	Declare any competing interests of review authors.	Page 25, line 3
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Appendix 2

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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Primary Subject Heading:	Health services research
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Keywords:	Public health < INFECTIOUS DISEASES, GENERAL MEDICINE (see Internal Medicine), PUBLIC HEALTH

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Definitions and measurement of health literacy in health and medicine research: a systematic review

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Abstract

Objectives: The way health literacy is understood (conceptualized) should be closely linked to how it is measured (operationalized). This study aimed to gain insights into how health literacy is defined and measured in current health literacy research and examine the relationship between health literacy definitions and instruments.

Design: Systematic review in accordance with the PRISMA statement.

Data sources: The Medline, PsycINFO, ERIC, and CINAHL databases were searched for articles published during two randomly selected months (March and October) in 2019.

Eligibility criteria: We included articles with a quantitative design that measured health literacy, were peer-reviewed and original, were published in the English language and included a study population older than 16 years.

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Data extraction and synthesis: Six researchers screened the articles for eligibility and extracted the data independently. All health literacy definitions and instruments were considered in relation to Category 1 (describing basic reading and writing skills, disease-specific knowledge, and practical skills) and Category 2 (social health literacy competence and the ability to interpret and critically assess health information). The categories were inspired by Nutbeam's descriptions of the different health literacy levels.

Results: 120 articles were included in the review: 60 within public health and 60 within clinical health. The majority of the articles (n=77) utilized instruments from Category 1. In total, 79 of the studies provided a health literacy definition; of these, 71 were in Category 2 and 8 in Category 1. In almost half of the studies (n=38), health literacy was defined in a broad perspective (Category 2), but measured with a more narrow focus (Category 1).

Conclusion: Due to the high degree of inconsistency between health literacy definitions and instruments in current health literacy research, there is a risk of missing important information about health literacy considered be important to the initial understanding of the concept recognised in the studies.

PROSPERO registration number: CRD42020179699.

Strengths and limitations of this study

- To our knowledge, this is the first systematic review to investigate connections between health literacy definitions and instruments used in current health literacy research.
- The health literacy definitions and instruments were categorized based on a health literacy model described by Nutbeam.

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- The initial plan to assess health literacy definitions and instruments for three categories was changed to two, due to difficulties in distinguishing between two of the categories.
- Searches were limited to two randomly selected months.

Word count: 5351

Introduction

Health literacy is usually understood as cognitive and social skills that determine the motivation and ability to understand and use health information,[1] and adequate health literacy is seen as a prerequisite for healthy behaviours. Researchers have increasingly worked to identify challenges associated with health literacy and investigate the role they play in an individual's ability to comprehend self-care information and its relationship to health outcomes.[2] Empirical studies have reported that low health literacy is associated with poor health-related outcomes, such as high hospital admission rates,[3-4] low participation in preventive activities,[5-9] poor self-management of chronic conditions,[10] poor disease outcomes, and high mortality.[11-13]

The concept of health literacy emerged in the 1970s when health education was viewed as social policy.[14] A topic-specific query in the PubMed tools reveals a recent exponential growth of articles about health literacy, with 129 references between 1986 and 1990 increasing to more than 8000 in the past five years. Today, health literacy is seen as a global goal for enhancing health promotion through improved education and communication strategies to improve health outcomes.[15]

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Health literacy is defined in numerous ways.[14, 16-18] In a systematic review by Sorensen et al.,[14] 17 different definitions and 12 conceptual models were reported on health literacy. Another systematic review found that 34 research articles between 2010 and 2015 had an explicit objective to define the concept of health literacy.[17] Moreover, the literature has reported the use and development of more than 150 health literacy instruments over the last decade.[19-21] Traditionally, health literacy approaches have focused on individual skill deficits and health education based on the communication of factual information regarding health risks as well as how to use the health system,[22] with the majority of health literacy research having used instruments measuring reading and numeric skills.[23] However, in recent years, more multi-dimensional perspectives and instruments measuring health literacy have been introduced, such as the Health Literacy Questionnaire[24] and the Health Literacy Survey European Questionnaire 47.[25]

The current and sometimes confusing use of various and inconsistent interpretations of health literacy is a challenge for the development of valid and reliable measurements.[16] In 2000, Nutbeam proposed a health literacy model that is now widely cited in the health literacy literature and is by many health literacy researchers seen as useful in analysing health literacy abilities required in various health situations. [22] According to Nutbeam, health literacy contains three different levels, progressing from basic skills in reading and writing (functional health literacy), to the ability to derive meaning from different forms of communication and apply new information to changing situations (interactive health literacy) and to more advanced cognitive skills which, together with social skills, can be applied to critically analyse information and to achieve policy and organizational changes (critical health literacy).

Different understandings of health literacy and different measurement tools may be useful as they complement each other and provide different perspectives. However, the way health literacy is understood (conceptualized) should be closely linked to how it is measured

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(operationalized) in each study context.[23] Ngyen et al. has described this as a "conceptual stumbling block" that needs to be resolved for the field to progress.[23] A first useful step for addressing this might be to systematically explore how it appears in current health literacy research. Hence, by performing a systematic review, our aim is to gain insights into how health literacy is defined and measured in current health literacy research. In particular, we will examine the relationship between health literacy definitions and instruments. This review may increase our understanding of potential conceptual and methodological challenges or gaps that need to be addressed in future research.

Methods

This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement[26] and registered in PROSPERO (No. CRD42020179699): https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=179699. The review was designed with a time frame limited to two randomly selected months in 2019 (March and October). Due to a high number of health literacy articles published every year, random selection was chosen in order to reflect current health literacy research. To ensure that March and October were not unique in terms of the number of articles published, we performed the same search strategy using the same databases for one other randomly selected month in 2019. This search yielded nearly the same number of articles.

Search Strategy

Systematic literature searches were conducted in collaboration with a trained librarian (3.4.2020). The Medline, PsycINFO, ERIC, and CINAHL databases were searched for the term "health literacy". For all databases except for ERIC (where this was not possible), the

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search was automatically restricted to two randomly selected months in 2019: March and October. Citations in ERIC were manually assessed for articles published in March 2019 and October 2019.

The search yielded a total number of 1038 citations. Endnote™ Version X9 was used to manage the generated research articles. After removing duplicates, 951 citations remained. All records not published for the first time in 2019 were removed, leaving 414 articles for screening. (See Appendix 1 for the search history).

Selection Criteria

The inclusion and exclusion criteria were developed a priori. The 414 published articles were distributed among six researchers (KHU, AKW, MHA, CRB, SH, MHL), who worked in pairs. The articles were included if they fulfilled the following inclusion criteria: 1) "health literacy" was mentioned in the title or abstract, 2) the article was peer reviewed, 3) the research was original, 4) it used a quantitative design, 5) it was published in the English language, and 6) the study population included individuals older than 16 years of age.

To reduce selection bias, the researchers independently screened the articles for eligibility according to the criteria. They then met in pairs to compare their results, resolve any conflicting opinions, and decide whether to include each article. Conflicting opinions in pairs were presented and solved by the whole team.

Data Extraction (Selection and Coding)

To achieve consistency in the data extracted from the included articles, an Excel spreadsheet was created. The initial question in this spreadsheet was whether health literacy was measured. If the answer was no, the article was excluded. The data extracted from the articles

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included information about the study design and context, such as country of origin and whether the study was conducted in a public health or clinical health setting. For clinical studies that included participants with health problems, the different types of diseases were categorized according to the International Statistical Classification of Diseases and Related Health Problems (ICD, WHO, 2010).[27] For the public health studies, we categorized type of study populations (Table 1). Furthermore, the data included information about whether the instruments were generic or disease specific, whether it was used in combination with other health literacy instruments or if a reference was provided for the instrument.

The published articles were analysed with respect to which health literacy definitions and instruments were used. Before the data extraction, a pilot assessment of three studies was performed to determine the feasibility of the data extraction tool. The initial plan was to assess health literacy definitions and instruments for three categories: 1) functional, 2) interactive, and 3) critical health literacy, inspired by Nutbeam' s description of different health literacy levels.[22] As a result of the pilot testing, the number of categories was changed due to difficulties in distinguishing between the two latter levels. All health literacy definitions and health literacy instruments were therefore considered in relation to two categories: Category 1 included definitions and instruments describing basic reading and writing skills, disease-specific knowledge, and practical skill competencies needed to function in everyday situations; and Category 2 included definitions and instruments that also described health literacy as skills to communicate and interact with healthcare providers as well as the ability to interpret and critically analyse health information (Appendix 2 illustrates the coding for the data extractions).

The researchers first extracted data and considered independently the definitions and instruments related to the two categories. They then met in pairs to compare their results and

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resolve any conflicting opinions. The remaining conflicting opinions within pairs were presented and solved by the whole team in a group meeting.

Patient and public involvement

No patient involved.

Results

The search yielded 1038 articles in total (see the Appendix for the search history). After removing duplicates, 951 articles remained. Of the 163 articles that were read in full text, one was a duplicate, and 39 were excluded because they did not measure health literacy. Ultimately, 120 articles were included (see Figure 1).

Figure 1 about here

Characteristics of Included Studies

Sixty articles described studies with a clinical health focus, and 60 were conducted in public health settings. The clinical studies included a great variety of diseases, but the most frequent diseases were those in the circulatory system (n=11) and endocrine diseases (n=9) followed by mental illness (n=6) and cancer diseases (n=6). In the public health studies, the most frequent study group was the general population (n=19), looking into aspects such as vaccination programmes, oral care and mental health. Furthermore, a major part of the public health studies included students (n=13), mainly represented by health professional students. A

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number of 6 studies focused on parents/care givers, while 5 studies focused on the elderly population.

Table 1. Overview of study populations in clinical health studies and public health studies

Clinical health studies		**Public health studies**	
Populations classified by Diseases and Related Health Problems	n	*Type of study population*	n
Diseases in the circulatory system	11	General population	19
Endocrine diseases	9	Students	13
Mental illness	6	Parents/care givers	6
Cancer	6	Elderly people	5
Diseases in the respiratory system	4	Immigrants	3
Diseases in the urine- and genital organs	4	Workrelated populations	3
Infectious- and parasite diseases	3	Young adults	2
Diseases in the ear	2	Veterans	2
Diseases in the musculoskeletal system and connective tissue	2	Health personell	2
Diseases in the nervous system	1	Men	1
Diseases in the digestive system	1	Women	1
		Pet owners	1
Pregnancy, birth, postnatal period	1	Relatives of patients with cancer	1
Other: General chronic conditions (n=2), chronic pain (n=3) patients in primary care (n=2) next of kind (n=2), patients in specialist care (n=1)	10	Churchgoers	1
Total	60	**Total**	60

The majority of the studies utilized a cross-sectional design (n=90). Nine studies utilized a randomized controlled design, and 10 studies were quasi-experimental. Four of the studies used a longitudinal design while one was a case-control study. Figure 2 presents an overview of all study designs.

Figure 2 about here

As Figure 3 indicates, the majority of studies were conducted in North America (n=56) and Asia (n=31). The fewest studies were performed in Africa (n=4) and South America (n=3).

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Figure 3 about here

Health Literacy Definitions and Instruments

Out of the 120 included studies, 88 utilized generic health literacy instruments while 32 were context-specific. Eleven studies used a combincation of health literacy instruments. A number of 77 studies utilized instruments from Category 1 (describing basic reading and writing skills, disease-specific knowledge, and practical skills), whereas 43 instruments were from Category 2 (describing communication and interaction skills and/or the ability to interpret and critically analyse health information). The most frequently used health literacy instruments in Category 1 were Newest Vital Sign[28] (n=19) and Test of Functional Health Literacy [29] (n=13). The most frequently used health literacy instruments in Category 2 were The European Health Literacy Survey Questionnaire[30] and The eHealth Literacy Scale[31] (n=10).

Of the 120 articles, 79 provided a health literacy definition. Of these, 46 were public health studies and 33 clinical health studies. Only eight studies provided a definition addressed to Category 1. Five out of these eight studies focused on mental health challenges.

The most frequently used reference in Category 2 was Nutbeam.[22,32,33] The most frequently used reference in Category 1 was a definition provided by Jorm and colleagues.[34-35] Table 2 gives an overview of the text in the most cited health literacy definitions. Words in italics are examples of key words considered important in the decision to add the definition to Category 1 or Category 2. For instance, definitions that described health literacy with terms such as "knowledge", "skills" and "attitude" were referred to Category 1, while definitions with terms like "appraise" and "social skills" were referred to Category 2.

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Table 2. The most frequently used references for health literacy definitions in the included studies

Reference	Definition	Number of studies
Category 1		
Jorm (1997) Jorm (2012)	"knowledge about appropriate treatment options; and *attitudes* that facilitate recognition and treatment-seeking" "*knowledge and beliefs* about mental disorders which aid their recognition, management or prevention"	5
American Medical Association (AMA) (1999)	"The *constellation of skills, including the ability* to perform basic reading and numerical tasks required to function in the healthcare environment"	1
Category 2		
Nutbeam (1998, 2000, 2015)	"The *cognitive and social skills* which determine the motivation and ability of individuals to gain access to understand and use information in ways which promote and maintain good health."	19
Sørensen (2012)	"Health literacy entails people's knowledge, motivation and competences to access*, understand*, *appraise*, and *apply* health information in order to make judgments and take decisions in everyday life concerning healthcare, disease prevention and health promotion to maintain or improve quality of life during the life course."	14
The Institute of Medicine (2004)	"The individuals' capacity to *obtain, process* and understand basic health information and services needed to make appropriate health decisions."	11

Regarding the connection between health literacy definitions and instruments in the 79 relevant studies, 41 articles used health literacy definitions and instruments characterized at the same level. Thirty-three of these were in Category 2, and 8 were in Category 1 (see Figure 4 and Table 3).

Figure 4 about here

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Table 3. Studies categorized with mutual levels of health literacy definitions and instruments (n=41)

Study reference	Study context	Design	Health literacy definition reference	Cat	Health literacy instrument	Cat.
Category 1						
Cehuen Neto et al., 2019	Clinical	Cross-sectional	American Medical Association (AMA) ad hoc committee on health, 1999	1	Short Test of Functional Health Literacy in Adults (Parker, 1995)	1
Clough et al., 2019	Public health	RCT	Jorm et al., 1997	1	Mental Health Literacy Scale (O'Connor and Casey, 2015)	1
Clough et al., 2019	Public health	Cross-sectional	Jorm et al. 1997	1	Mental Health Literacy Scale (O'Connor & Casey, 2015)	1
Joncho et al., 2019	Clinical	Cross-sectional	Jorm et al., 1997	1	Developed within the study context: Mental Health Literacy Questionnaire	1
Lichtveld et al, 2019	Public health	Cross-sectional	Finn and O'Fallon, 2019 Gray, 2018	1	Developing and validating a questionnaire measuring environmental health literacy: Measuring media-specific knowledge, attitudes, and behaviours	1
Nye et al., 2019	Public health	Longitudinal	U.S. Department of Health and Human Services, 2010	1	The Upper Peninsula Oral Health Assessment Survey	1
Waldmann et al.,2020	Clinical	Cross-sectional	Jorm et al., 2012	1	Mental Health Knowledge Schedule. Depression Literacy Scale (Wilson, 2015)	1
Williston et al., 2020	Public health	Longitudinal	Jorm et al., 1997	1	Mental Health Literacy Scale (O'Connor and Casey, 2015)	1
Category 2						
An et al., 2019	Public health	Cross-sectional	Nutbeam, 2000	2	The Health Literacy Scale (Chew, 2004)	2
Azizi et al., 2019	Public health	Mixed method	Nutbeam, 2000	2	Health Literacy Scale for Workers (HELSW)	2
Brandsetter et al., 2019	Public health	Cross-sectional	Sørensen, 2012	2	The European Health Literacy Survey (HLS-EU) (Sørensen et al., 2013)	2
Degan et al., 2019	Clinical	Cross-sectional	World Health Organization, 2009	2	The Health Literacy Questionnaire (Osborn et al., 2013)	2
Eo et al., 2019	Public health	Cross-sectional	Nutbeam, 2000	2	The Health Literacy Assessment Scale for Asian Immigrant Women (And and Yang, 2015)	2
Ernsting et al., 2019	Clinical	Cross-sectional	Sørenesen, 2012	2	European Health Literacy Survey Questionnaire - short form (Sørensen et al., 2013) and The eHealth Literacy Scale (Norman, 2006)	2
Erunal et al., 2019	Public health	RCT	Nutbeam, 2000	2	Turkish Health Literacy Scale -32 (Okyay et al., 2016)	2
Evans et al., 2019	Public health	Cross-sectional	Sørenesen, 2012	2	European Health Literacy Survey Questionnaire (Sørensen et al., 2013)	2
Fernandez Gutierrez et al., 2019	Public health	Quasi-experimental	Nutbeam, 2000	2	European Health Literacy Survey Questionnaire -HLS EU-16 (Sørensen et al., 2013) Health Literacy App Questionnaire + five practical tests	2
Goto et al., 2019	Public health	Cross-sectional	Nutbeam, 1998	2	The Functional Communicative and Critical HL Tool (Ishikawa et al., 2008)	2
Guclu et al., 2019	Clinical	Cross-sectional	World Health Organization, 2009	2	European Health Literacy Survey Questionnaire (Sørensen et al., 2013)	2
Gûner et al., 2019	Public health	Cross-sectional	Nutbeam, 2000	2	Questionnaire developed within the study context (consisting of 21 questions about HL knowledge, practice, and attitudes)	2
Hu et al., 2019	Public health	Cross-sectional	Nutbeam, 2000	2	Health Literacy Management Scale (Jordan, 2013)	2

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Indino et al 2019	Public health	Cross-sectional	Nutbeam, 2015	2	The Functional Communicative and Critical Health Literacy Tool (Ishikawa et al., 2008)	2
Kaper et. al, 2019	Public health	Quasi-experimental	Kwan, 2006	2	Questionnaire developed within the study context by combining subscales from other questionnaires	2
Kobayashi et al., 2019	Public health	Mixed method	Sørensen, 2012	2	European Health Literacy Survey Questionnaire (Sørensen et al., 2013)	2
Lim et al., 2019	Clinical	Cross-sectional	Osborne, 2013	2	Health Literacy Questionnaire (Osborne et al., 20139	2
Logullo et al., 2019	Public health	Cross-sectional	Sørensen, 2012	2	Short Assessment of Health Literacy for Portuguese Speaking Adults (SAHL-PA) (Apolonario et al., 2012)	2
Lu et al., 2020	Clinical	Cross-sectional	Sørensen et al., 2012	2	European Health Literacy Survey Questionnaire -HLS EU-16 (Sørensen et al., 2013)	2
Lu et al., 2019	Clinical	Cross-sectional	Sørensen, 2012	2	European Health Literacy Survey Questionnaire -HLS EU-16 (Sørensen et al., 2013)	2
Ma et al., 2019	Public health	Cross-sectional	Norman et al., 2006	2	The eHealth Literacy Scale (Norman, 2006)	2
Nokes et al, 2019	Clinical	Quasi-experimental	Norman et al., 2006 and 2011	2	The eHealth Literacy Scale (Norman, 2006)	2
Oh et al., 2019	Public health	Cross-sectional	World Health Organization, 2009	2	The eHealth Literacy Scale (Norman, 2006)	2
Oliffe et al, 2019	Public health	Cross-sectional	Peerson and Saunders, 2009, O'Brian et al., 2005	2	The Health Literacy Questionnaire (Osborn et al., 2013)	2
Pobhirun and Pinitsoontorn, 2019	Public health	Cross-sectional	Nutbeam, 1998	2	A questionnaire developed within the study context including 6 dimensions: cognitive skills, access, communication skills, self-management, media literacy and decision-making skills	2
Rababah et al, 2019	Public health	Cross-sectional	World Health Organization, 2013, Sørensen, 2012	2	The Health Literacy Questionnaire (Osborn et al., 2013)	2
Ruëgg and Abel, 2019	Public health	Cross-sectional	Selden et al., 2000	2	Short survey tool for public health and health promotion research (Abel, 2014)	2
Solhjoo Niloofar et al., 2019	Public health	Mixed method	Sørensen, 2012	2	The eHealth Literacy Scale (Norman, 2006)	2
Strømer et al., 2019	Clinical	Cross-sectional	World Health Organization, 1998	2	The Health Literacy Questionnaire (Osborn et al., 2013)	2
Ulysal et al., 2019	Public health	Cross-sectional	Sørensen, 2012	2	European Health Literacy Survey Questionnaire (Sørensen et al., 2013)	2
Wang et al., 2019	Clinical	Cross-sectional	Kutner et al., 2003	2	Chinese Health Literacy Scale for Diabetes (Leung, 2013)	2
Wang et al., 2019	Clinical	Cross-sectional	Nutbeam, 2009	2	Diabetes Health Literacy Scale (Lee, 2018)	2
Zhang et al., 2019	Clinical	Cross-sectional	Sørensen, 2012	2	Heart Failure-specific Health Literacy scale (HFHL) (Matsuka et al., 2018)	2

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In the remaining 38 studies, there was a disconnect between levels of health literacy definitions and instruments. In all of these, health literacy definitions were from Category 2, and all instruments were in Category 1 (see Table 4).

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Table. 4 Studies categorized with conflicting levels of health literacy definitions and instruments (n=38)

Study reference	Study context	Design	Health literacy definition reference	Cat.	Health literacy instrument	Cat.
Anderson et al., 2019	Clinical	Cross-sectional	Defined within the study context (no reference provided)	2	Short Test of Functional Health Literacy in Adults (S-TOFHLA) (Parker, 1995)	1
Avci et al.,2019	Public health	Cross-sectional	US Department of Health and Human Services, 2000	2	Newest Vital Sign (NVS) (Weiss et al., 2005) and Rapid Estimate of Adult Literacy in Medicine (Davis, 1993)	1
Bonaccorsi et al., 2019	Public health	Cross-sectional	Sørensen, 2012	2	Newest Vital Sign (Weiss et al., 2005)	1
Bonaccorsi et al., 2019 II	Public health	Cross-sectional	Sørensen, 2012	2	Newest Vital Sign (Weiss et al., 2005)	1
Carducci et al., 2019	Public health	Cross-sectional	Ratzan and Parker, 2000 Nutbeam, 2000	2	Test of Functional Health Literacy in Adults (TOFHLA) (Parker, 1995)	1
Chen et al., 2019	Clinical	Cross-sectional	Nutbeam, 2000	2	Health Literacy Scale for Diabetes (Lee et al., 2016)	1
De Melo et al., 2019	Clinical	Cross-sectional	Cavanaugh, 2011	2	Short Test of Functional Health Literacy in Adults (Parker, 1995)	1
Flynn et al., 2019	Public health	Cross-sectional	American Dental Association Council, 2015	2	Oral Health Literacy Adults Questionnaire (Sistani et al., 2014)	1
Gaikwad, 2019	Public health	Cross-secttional	Ratzan and Parker, 2000	2	Rapid Estimation of Adult Literacy in Dentistry 30-word version (Lee et al., 2007)	1
Gûner et al., 2019	Public health	Cross-sectional	Nutbeam, 2000	2	Developed within the context of the study	1
Han et al.,, 2019	Public health	Cross-sectional	Sørensen, 2012, Nutbeam, 2000	2	Assessment of Health Literacy in Cancer Screening (AHL-C) (Han et al., 2014)	1
Himes et al., 2019	Public health	Cross- sectional	Institute of Medicine, US, Committee HL, 2004	2	Chew et al.'s (2004) set of brief questions	1
Irvin et al., 2019	Public health	Cross-sectional	Peters et al., 2012	2	The Water Environmental Literacy Level Scale (WELLS) (Irvin et al., 2019)	1
Kaur et al., 2019	Public health	RCT	Healthy People, 2010 (Oral health)	2	TS-REALD Two-Stage Rapid Estimate of Adult Literacy (Stucky et al., 2011)	1
Kim et al., 2019	Public health	Cross-sectional	Ratzan and Parker, 2000	2	Three questions, each addressing oral, listening, and written literacies (no reference)	1
Kim et al., 2019	Public health	Quasi-experimental	Joint Committee on National Education Standards, 1995	2	Knowledge questions developed in the study context	1
Kim et al, 2019	Clinical	RCT	Institute of Medicine, US, Committee HL, 2004	2	Short Test of Functional Health Literacy in Adults (Parker, 1995)	1
Kino et al., 2020	Public health	Cross-sectional	Inst of Medicine, 2004	2	Three indicators of HL (Haun et al., 2014)	1
Lin et al., 2019	Public health	Quasi-experimental	Nutbeam, 2000	2	Mandarin Health Literacy Scale (Lee et al., 2012)	1
Lindahl et al., 2020	Clinical	Cross-sectional	Nutbeam and Kickbusch, 2000	2	The three-item Brief Health Literacy Screen (BHLS) (Cavanaugh et al., 2015)	1

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Mackert et al., 2019	Public health	Cross-sectional	Berkman et al., 2010	2	Newest Vital Sign (NVS) (Weiss et al., 2005)	1
Mayer et al., 2019	Clinical	Cross-sectional	Nutbeam, 1998	2	Developed within the context of the study	1
Meyers et al., 2019	Clinical	Cross-sectional	Institute of Medicine, 2004	2	Newest Vital Sign (Weiss et al., 2005)	1
Miranda, 2019	Clinical	Cross-sectional	Sørensen et al., 2012	2	Rapid Estimate of Adult Literacy Dutch version (REALM-D) (Davis, 1993)	1
Mock et al., 2019	Clinical	Cross-sectional	Institution of Medicine, 2004	2	Single Item Literacy Screening (SILS) (Morris, 2006), Short Test of Functional Health Literacy in Adults (S-TOFHLA) (Parker, 1995)	1
Mora-Pinzon et al., 2019	Clinical	Cross-sectional	Institute of Medicine, 2004	2	Short Test of Functional Health Literacy in Adults (S-TOFHLA) (Parker, 1995) Rapid Estimate of Adult Literacy (REALM-D) (Davis, 1993	1
Noback et al., 2019	Clinical	Cross-sectional	Paasche-Orlow et al., 2007, Berkman et al., 2011, Pendlimari et al., 2012	2	Newest Vital Sign (NVS) (Weiss et al., 2005) Musculoskeletal specific literacy survey (LiMP questionnaire) developed within the study context	1
O'Conor et al., 2019	Clincial	Cross-sectional	Institute of Medicine Health Literacy, 2004	2	Short Test of Functional Health Literacy in Adults (S-TOFHLA) (Parker, 1995)	1
Penaloza et al, 2019	Clinical	Cross-sectional	Nutbeam, 1998 and Institute of Medicine, 2004	2	Short Assessment of Health Literacy Spanish (SAHL-S) (Lee et al., 2006)	1
Rafferty et al., 2019	Clinical	Cross-sectional	Institute of Medicine, 2004 and Sørensen, 2012	2	Questions focusing on health advice and information-seeking skills, oral literacy, and print literacy	1
Sany et al., 2019	Clinical	Quasi-experimental	United States Health Resources and Services Administration	2	Test of Functional Health Literacy in Adults (TOFHLA) (Parker, 1995)	1
Schrivner et al., 2019	Clinical	Cross-sectional	Nutbeam, 2008	2	Three questions assessing health literacy (no reference)	1
Tucker et al., 2019	Public health	Quasi-experimental	Ratzan and Parker, 2000	2	Newest Vital Sign (NVS) (Weiss et al., 2005)	1
Van Wormer et al., 2019	Public health	Cross-sectional	Institute of Medicine, 2004	2	Oral Health Literacy in Adults Questionnaire (Sistani et al., 2013)	1
Weaver et al., 2019	Public health	Cross- sectional	Ratzan and Parker, 2000 Nutbeam, 2000	2	Rapid Estimate of Adult Literacy (REALM) (Davis, 1993)	1
Zhong et al., 2020	Clinical	Cross-sectional	Kindig, 2004	2	Newest Vital Sign (NVS) (Weiss et al., 2005)	1
Williams et al, 2019	Clinical	Cross-sectional	Institute of Medicine, 2004	2	Newest Vital Sign (NVS) (Weiss et al., 2005)	1
Winokur et al., 2019	Clincial	Cross-sectional	Morrison et al., 2013	2	Newest Vital Sign (NVS) (Weiss et al., 2005)	1

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Discussion

This systematic review aimed to gain insights into how current research defines and measures health literacy and, in particular, whether studies consistently used definitions and instruments. The high number of articles published in the defined timeframe shows that health literacy is of high research interest in both public health and the clinical field in large parts of the world.

We found a large variety of instruments used, and the majority of the included studies (79 out of 120) presented a health literacy definition as part of the study's theoretical background. However, there seems to be an inconsistency between the definitions and the instruments in a significant number of the studies. In nearly half of the studies, health literacy was defined in a broad perspective (including aspects such as social health literacy competence and the ability to process and appraise health information) while using instruments with a more narrow focus (measuring basic skills and knowledge). As a result, almost half of the articles in our review lacked data on the participants` ability to critically appraise health information and their social health literacy competence despite the fact that the authors had stated such aspects to be health literacy. This concern has previously been addressed. Numerous systematic reviews have reported on the diversity of understandings of health literacy and the various use of instruments not aligned to the definitions in current research.[16-21] However, the current study is, to the best of our knowledge, the first review to systematically investigate the relationship between health literacy definitions and instruments from the perspective of a health literacy model.

The use of instruments that focused on functional health literacy (addressed to Category 1) such as Newest Vital Sign and Test of Functional Health Literacy, were predominant in our review.[28,29] This finding corresponds with previous findings.[19,20] The widespread use of

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Newest Vital Sign and Test of Functional Health Literacy is justified by the fact that they are screening tools—that is, they are quick, available in an ever-increasing number of languages, and can be adapted to different settings.[36] However, a broad range of skills and tasks covering functional, interactive, and critical domains needs to be included in measures of health literacy in order to capture the health literacy definitions used.[32] In recent years, an increasing number of multi-dimensional instruments have been developed. In our review, The European Health Literacy Survey Questionnaire and The eHealth Literacy Scale, both addressed to Category 2, were the multi-dimensional instruments most often used .[30,31] In light of the increased focus on a more multi-dimensional perspective of health literacy, and the fact that the current review only includes studies from 2019, one could expect a higher use of multidimensional instruments. Instead, they represented only 43 out of 120 instruments.

Clearly, our study highlights the discrepancy between definitions and measurements as well as the narrow focus of health literacy in a large amount of empirical research. An important question to raise is therefore how the findings from this specific study can guide future research strategies to overcome the identified inconsistency. In other words, as most health literacy researchers seem to base their research on a broad understanding of health literacy, what can be done to facilitate an increased use of broad measurements? A first step should be to make researchers aware about the existing mismatch in current research. Furthermore, it seems necessary to develop more instruments that can answer the research questions posed. Despite the high number of instruments, there still seems to be a need for questionnaires in the field of health literacy that capture more multidimensional dimensions besides the functional aspects.

Nutbeam' s description of three levels of health literacy inspired the categorization of definitions and instruments.[22] Initially, we planned to distinguish among the three levels. However, the line between Levels 2 and 3 was challenging to distinguish. Therefore, it was

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decided to merge the two latter categories. In a recent publication from 2020, Nutbeam has provided a more thorough description of the three levels with a more detailed explanation of how the levels should be understood.[37] Perhaps, these descriptions would have contributed to a clearer guidance in our work with the categorization. However, this material was not available at the time of our work and, in general, this situation illustrates the challenge of adapting a theoretical model into practise.

The interpretation of terms used in the definitions and instruments that guided the choice of category also represented some challenges. These were resolved through discussions both in pairs and as a research team. For instance, definitions describing health literacy using terms like "knowledge" and "beliefs" were included in Category 1 while definitions using terms such as "appraise" and "understand and process" were included in Category 2. An interesting finding is that, among the eight studies providing a health literacy definition from Category 1, the majority (n=5) were in the context of mental health. The predominant reference in these studies was the definition provided by Jorm and colleagues who, unlike others, defined health literacy as being linked to "beliefs" and "attitudes".[34,35] Whether these terms are more closely linked to mental health literacy challenges, compared to other more "physical" health literacy issues, is not clear but would be interesting for further investigation.

Another aspect worth noting is that we discovered that many publications did not cite the primary source of the health literacy definitions, but instead referred to secondary sources (other researchers presenting definitions of health literacy). Referring to the original sources should be the first choice and would perhaps make it easier for readers to recognize the definition's affiliation.

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The present study has some limitations. First, this study was designed to analyse and describe health literacy research in two randomly selected months. This period may not be representative of health literacy research in general. However, a large number of health literacy studies are published every year. A random selection can therefore give a good picture of health literacy research. Second, we did not conduct a quality assessment screening of the included studies. This was considered less relevant for the current study as the aim of the current study was to explore connections between health literacy definitions and instruments rather than to assess methodology. Furthermore, searches were limited to the English language only. It is possible that similar studies may have been published in languages other than English.

The current review only included quantitative measurements. However, qualitative approaches might provide valuable and more in-depth insights into the field. For future research, it would be interesting to also explore how qualitative research links health literacy definitions to the research questions posed.

Conclusion

There seems to be an inconsistency between the definitions and the instruments used in a significant part of current health literacy research. This situation raises the risk of missing information about health literacy that was considered to be important in the initial understanding of the concept recognised in the studies. This gap should be taken into consideration in future health literacy research. We hope our work contributes to making explicit where the problem might be rooted and that it can be useful in the discussion about strategies for moving forward to better align health literacy measurement with definitions of health literacy.

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Competing interest statement: No competing interests**.**

Ethics statements: Patient consent for publication not required.

Data availabillity: The data from the study will be made available upon request to the corresponding author.

Figure legends:

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Figure 1 Flow diagram shows the study selection process

Figure 2 Overview of study designs of included studies

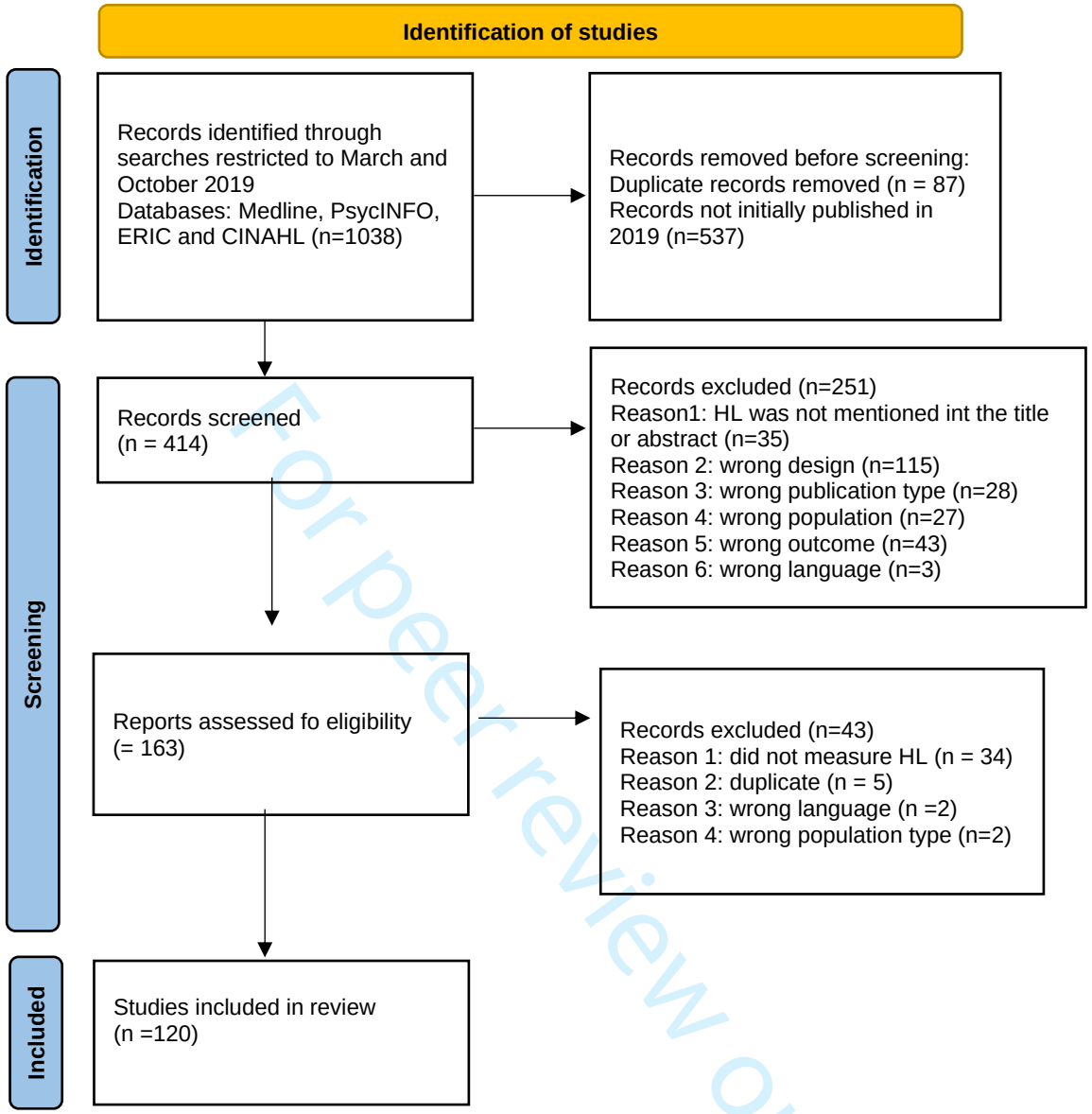
Figure 3 Studies' continent of origin

Figure 4 Studies categorized with the same level of HL definitions and HL instruments

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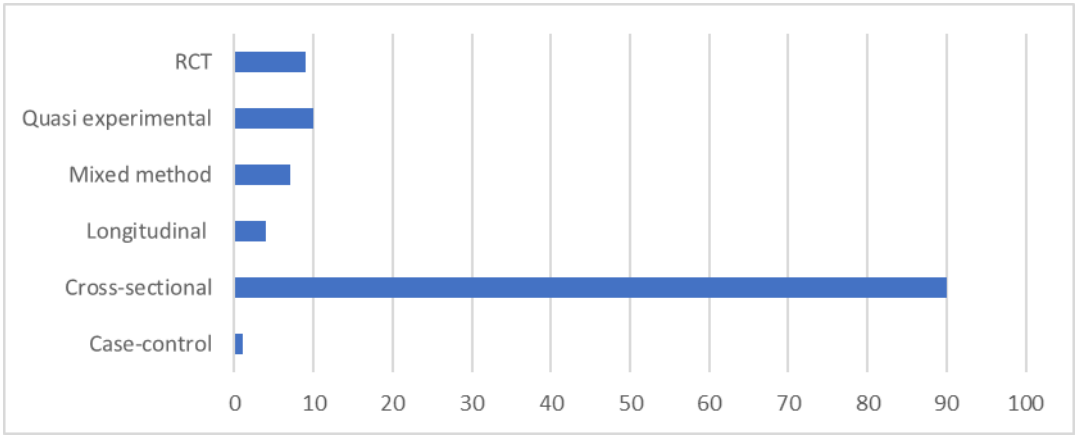
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Figure 1 Flow diagram shows the study selection process



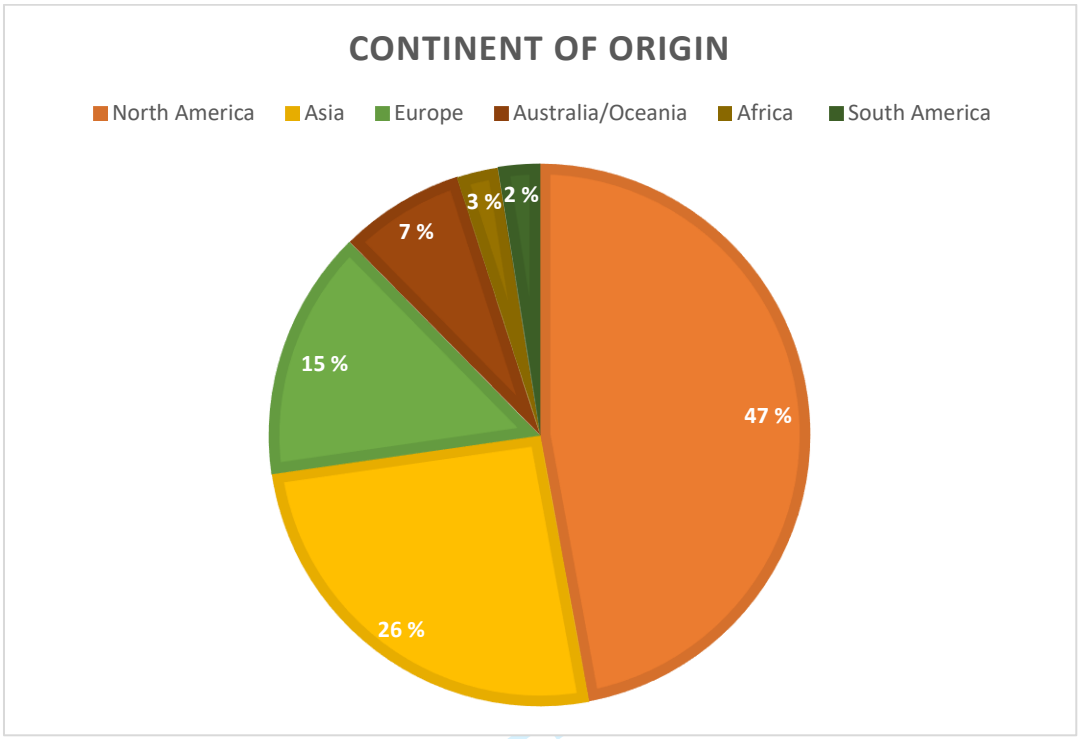
For peer review only

Figure 2. Overview of study designs of included studies



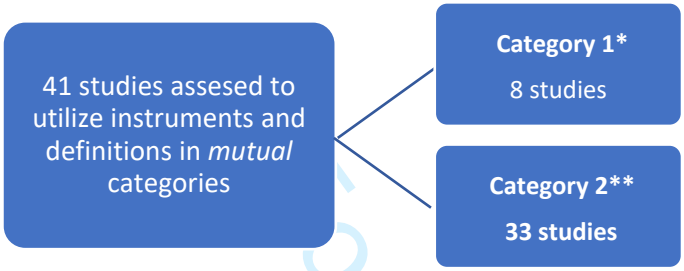
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Figure 3. Studies' continent of origin



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Figure 4. Studies categorized with the same level of HL definitions and HL instruments



**Category 1: basic reading and writing skills, disease-specific knowledge and practical skill competences*

***Category 2: including skills to communicate and interact with healthcare providers and the ability to interpret and critically analyse health information*

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Search history

Databases	Medline
Date	03.04.2020
Search history	1 Health Literacy/ (5393) 2 health literacy.ti,ab,kw. (8163) 3 1 or 2 (10010) 4 limit 3 to (dt=20191001-20191031 or rd=20191001-20191031) (324) 5 limit 3 to (dt=20190301-20190331 or rd=20190301-20190331) (210) 6 4 or 5 (532)
Number of hits	532
Comments	Limited to Entry Date (March + October 2019) from https://wkhealth.force.com/ovidsupport/s/article/Limit-by-date-range-in-Ovid

Databases	PsycINFO
Date	03.04.2020
Search history	1 exp health literacy/ (2991) 2 Health Literacy.ti,ab,id. (3744) 3 1 or 2 (4291) 4 limit 3 to up="20191001-20191031" (30) 5 limit 3 to up="20190301-20190331" (18) 6 4 or 5 (48)
Number of hits	48
Comments	Limited to Entry Date (March+ October 2019) from https://wkhealth.force.com/ovidsupport/s/article/Limit-by-date-range-in-Ovid

Databases	CINAHL with Full Text
Date	03.04.2020
Search history	S5 S3 AND S4 Search modes - Boolean/Phrase S4 EM 201903* or EM 201910* or RD 201903* or RD 201910* Search modes - Boolean/Phrase S3 S1 OR S2Search modes - Boolean/Phrase S2 "Health Literacy" Search modes - Boolean/Phrase S1 (MH "Health Literacy") Search modes - Boolean/Phrase
Number of hits	419

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Search history

Comments	Limited to Entry Date (March+ October 2019

Databases	ERIC
Date	03.04.2020
Search history	S2 "Health Literacy" Limiters - Date Published: 20190101-20191231 Search modes - Boolean/Phrase S1 "Health Literacy" Search modes - Boolean/Phrase
Number of hits	39
Comments	Searched for Publication Date - 2019 (All year)

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Appendix 2 Coding for the data extractions

1 .Does this study measure health literacy?	2. Perspective	3 .Design	4. Country of origin	5. Status population	6. Type of disease	7. Other disease
Yes No	Public health Clinical health care setting	RCT Quasi experimental Mixed method Longitudinal Cross-sectional Case-control		Healthy With a diagnose	Infectious- and parasite diseases Cancer Endocrine diseases Mental illness Diseases in the nervous system Diseases in the ear Diseases in the circulatory system Diseases in the respiratory system Diseases in the digestive system Diseases in the musculoskeletal system and connective tissue Diseases in the urine- and genital organs Pregnancy, birth, postnatal period	

8.Is HL defined?	9. Reference for HL defintion:	10. HL definition category
Yes No		Category 1 Category 2

11. HL instrument used	12. Reference for instrument	13. Provide reference	14. Type of HL instruemnt	15. HL instrument category
Name:	Yes No		Generic Disease specific	Category 1 Category 2

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 2, line 6
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 2, line 32
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 4-5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 5, line 44
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 6, line 56
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 6, line 23
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 7, line 3
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 7, line 3
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 6, line 56
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 6, line 56
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Not relevant
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Not relevant
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Not relevant
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Not relevant
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Not relevant
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Not relevant

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Not relevant
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Not relevant
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 8, line 38
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 9, line 12
Study characteristics	17	Cite each included study and present its characteristics.	Page 10, line 9
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Not relevant
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Not relevant
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Not relevant
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Not relevant
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Not relevant
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Not relevant
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Not relevant
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Not relevant
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 19, line 37
	23b	Discuss any limitations of the evidence included in the review.	Page 21, line 44
	23c	Discuss any limitations of the review processes used.	Page 21, line 44
	23d	Discuss implications of the results for practice, policy, and future research.	Page 22, line 24
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 3, line 23
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Page 3, line 23

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Page 7, line 55
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 24, line
Competing interests	26	Declare any competing interests of review authors.	Page 25, line 3
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Appendix 2

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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